Transcending the Linearity Bottleneck: A Physics-Informed Quadratic Transformer for Bearing Fault Diagnosis under High-Entropy Noise

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Abstract

Rolling element bearing fault diagnosis in industrial machinery is frequently hindered by severe environmental noise, variable operational loads, and the stochastic nature of real-world degradation. Conventional deep learning models, which predominantly rely on linear convolutions, often struggle to demodulate amplitude-modulated fault signatures under low Signal-to-Noise Ratios (SNR) and exhibit limited cross-domain generalization. To address these inherent limitations, this paper proposes PhyQ-TransNet, a physics-informed architecture that integrates Enhanced Quadratic Convolutions (EQC) with a Transformer Encoder. Unlike standard Convolutional Neural Networks (CNNs), the EQC module operates as a learnable non-linear demodulator, explicitly extracting energy-based fault features and recovering high-frequency resonant kinematics. To ensure robust validation, a rigorous, leakage-free evaluation protocol was implemented across five diverse benchmarks: CWRU, SDUST, IMS, Paderborn, and NREL. Experimental results demonstrate the exceptional resilience of the proposed model, achieving 99.98% accuracy on the IMS run-to-failure dataset, 99.03% on real-world Paderborn fatigue damages with strong calibration (ECE=0.0047), and sustaining 100% accuracy on the 8-channel NREL wind turbine gearbox dataset under extreme -10 dB noise conditions. These findings indicate that PhyQ-TransNet provides a highly reliable and noise-robust framework for industrial condition monitoring, demonstrating substantial improvements over contemporary baselines.

Keywords: Bearing fault diagnosis, Physics-informed deep learning, Quadratic convolution, Transformer, Noise robustness, Domain generalization

1. Introduction

The relentless global transition toward carbon-neutral energy infrastructures has precipitated a rapid expansion in wind power capacity, particularly in offshore environments where wind resources are abundant but operational conditions are notoriously harsh. Wind Turbines (WTs), as the cornerstone of this transition, operate under stochastic loads, corrosive humidity, and variable speeds, subjecting critical subsystems to immense mechanical stress [1]. Among these subsystems, the drivetrain—encompassing the main bearing, gearbox, and generator—remains the leading source of turbine downtime. Statistical reliability analyses consistently identify rolling element bearings within the gearbox as the weakest link, where premature failures drive up Operation and Maintenance (O&M) costs and disrupt grid stability [2, 3]. Consequently, developing robust Fault Diagnosis (FD) frameworks capable of early defect detection is imperative for ensuring the reliability of modern wind farms.

The central challenge in bearing fault diagnosis is the extraction of weak, repetitive transient impulses buried under heavy background noise and non-stationary interferences [4]. Historically, Signal Processing (SP) techniques served as the primary diagnostic tools. Methods rooted in Fourier theory proved effective for stationary signals but faltered under time-varying operational conditions. To address non-stationarity, time–frequency analysis tools [5–7] and adaptive decomposition algorithms [8–10] were introduced. Parallel to SP methods, Blind Deconvolution (BD) algorithms emerged to recover impulsive signatures by optimizing statistical norms [11–15]. Despite their theoretical elegance, these traditional methods rely heavily on expert knowledge for hyperparameter tuning, remain computationally expensive, and lack the scalability required for automated industrial monitoring pipelines [16, 17].

The advent of Deep Learning (DL) revolutionized fault diagnosis by enabling end-to-end feature learning. Convolutional Neural Networks (CNNs) became the de facto standard, demonstrating that raw vibration signals could be directly processed to suppress high-frequency noise [18, 19]. Recently, research has pivoted toward highly complex paradigms to handle domain shifts and limited data. For instance, Generative models and Diffusion processes [20, 21] have been employed for data synthesis, while other studies advocate for Signal-to-Image (S2I) transformations coupled with Graph Convolutional Networks (GCNs) or visual architectures [22, 23]. Furthermore, integrating physical priors into neural networks has emerged as a prominent frontier [24, 25]. However, these cutting-edge 2025–2026 approaches face critical bottlenecks in real-time industrial deployment: (1) **Computational Overhead:** GCNs incur an exponential $O(N^2)$ algorithmic complexity, and S2I transformations severely increase latency, making them impractical for real-time edge deployment. (2) **Training Instability:** Genera-

tive models (e.g., GANs) risk feature hallucination and demand massive computational resources. (3) **Superficial Physics Integration:** Current physics-guided models often relegate physical constraints to the loss function rather than embedding them structurally, failing to effectively demodulate the nonlinear energy signatures of naturally degraded bearings under severe noise (e.g., -10 dB SNR).

To bridge the gap between theoretical representation learning and industrial reliability, this study identifies three fundamental limitations in contemporary FD frameworks:

1. The Linearity Bottleneck: Standard CNNs rely on linear cross-correlation operations. Recent theoretical insights [26, 27] suggest that linear operations are inherently suboptimal for characterizing bearing faults, which manifest as nonlinear, amplitude-modulated (AM) signals. Linear kernels act as band-pass filters but lack the mathematical capacity to perform the necessary *envelope demodulation*. Recent breakthroughs, such as the Multi-scale Quadratic CNN [28], demonstrate that replacing linear kernels with second-order Volterra filters provides a physics-informed pathway to capture energy fluctuations. However, scaling these networks for global receptive fields remains a formidable challenge.

2. Limited Global Context: Most CNN-based models exhibit a "locality bias," neglecting the long-range temporal dependencies and cyclostationarity intrinsic to vibration sequences. While Recurrent Neural Networks (RNNs) [29, 30] address this, their sequential processing hinders parallelization. Transformer architectures offer superior global context modeling, yet their efficient integration into 1D time-series diagnostic models without massive computational overhead remains underexplored.

3. Noise Sensitivity in Attention: While attention mechanisms highlight salient features [31, 32], standard modules calculate weights based on raw feature correlations. Under severe signal submergence, these modules often attend to high-energy Gaussian noise rather than the sparse fault impulses, leading to feature washout.

Addressing these challenges, we propose a novel, physics-aware heterogeneous architecture termed **PhyQ-TransNet** (Physics-Informed Quadratic Transformer Network). Unlike conventional black-box models or computationally heavy GCNs, our framework is designed to mathematically mirror the physical process of fault demodulation while maintaining strict computational frugality. The architecture integrates a dual-stream pipeline: a Deep Quadratic Signal Stream utilizing Enhanced Quadratic Convolutions (EQC) and a Transformer Encoder, synergized with a parallel Physics-Informed Statistical Stream to anchor the decision boundaries under extreme uncertainty.

The primary contributions of this work are summarized as follows:

1. **Physics-Informed Demodulation via Quadratic Kernels:** We introduce a 1D quadratic convolutional backbone that mathematically emulates the square-law demodulation process. This enables the network to actively extract amplitude-modulated fault energy envelopes directly from raw time-series data, significantly outperforming linear CNNs in morphological feature extraction.
2. **Hybrid Dual-Stream Architecture:** We propose a novel integration of high-dimensional deep representations (local EQC features and global Transformer contexts) with robust, low-dimensional statistical priors. This dual-pathway design is dynamically regulated by an Adaptive Gated Fusion (AGF) module, explicitly preventing performance collapse under severe noise.
3. **Comprehensive Generalization across Five Benchmarks:** Moving beyond the standard CWRU baseline, we validate the framework's scalability and load-invariant generalization on four additional rigorous datasets: SDUST (variable defect severity), IMS (run-to-failure), Paderborn (real fatigue damages), and NREL (utility-scale wind turbine gearbox).
4. **Exceptional Noise Robustness and SOTA Reliability:** Extensive experiments demonstrate state-of-the-art diagnostic resilience. Operating directly on 1D signals without heavy 2D transformations, the proposed model achieves **99.03%** accuracy on real-world fatigue faults (Paderborn) and sustains **99.54%** accuracy on the SDUST dataset via an Ensemble Voting strategy, substantially outperforming recent 2025–2026 baseline architectures while ensuring real-time inference suitability.

2. Related Work

2.1. Deep Learning Advancements: From CNNs to Generative Models

Over the past decade, Deep Learning (DL) has fundamentally revolutionized industrial fault diagnosis by automating the feature engineering pipeline. The seminal introduction of the Wide Deep Convolutional Neural Network (WDCNN) by Zhang et al. [18] demonstrated that wide temporal kernels could effectively suppress high-frequency noise. Following this trajectory, multi-scale architectures emerged to capture degradation signatures at varying granularities [11, 33]. To specifically address heavy industrial noise, Zhao et al. [19] proposed the Deep Residual Shrinkage Network (DRSN), a landmark architecture that integrated soft-thresholding to adaptively eliminate noise-related activations.

In the most recent academic landscape (2025–2026), the research focus has aggressively shifted toward tackling data scarcity and severe domain shifts through highly complex generative and distributed paradigms. For example, Guo et al. [20] introduced a condition-guided Diffusion Model (DiffUCD) to synthesize unseen fault samples under unknown working conditions, while Zhang et al. [21] developed an improved CSGAN-GP framework with dynamic filtering to stabilize small-sample training. Concurrently, Federated Learning frameworks, such as the open-set approach by Xu et al. [34], have been proposed to leverage distributed industrial telemetry while preserving corporate privacy.

However, these generative and federated methods often incur prohibitive computational overhead and risk "feature hallucina-

tion"—a fatal flaw in safety-critical prognostics. In stark contrast to these data-driven approaches that attempt to synthesize missing information through unstable adversarial training, our work advocates for a strictly discriminative, physics-anchored backbone. This approach maximizes structural information extraction directly from the raw, noisy signals, entirely bypassing the computational burden of generative sampling.

2.2. Physics-Informed Perception & Signal Processing Integration

A major paradigm shift towards "Physics-Informed" Deep Learning seeks to structurally embed signal processing priors directly into neural architectures. Early attempts constrained convolutional filters to predefined mathematical basis functions, such as WaveletKernelNets [6, 7]. More recently, researchers have focused on transforming 1D vibration sequences into 2D topological representations to leverage visual architectures. For instance, Asy'ari et al. [23] utilized Bispectrum image analysis for wind turbine gearboxes, and Dong et al. [35] employed Threshold-Free Recurrence Plots (TFRP) to capture nonlinear dynamics. While diagnostically effective, these Signal-to-Image (S2I) transformations significantly inflate memory utilization and preprocessing latency.

Another prominent frontier involves unfolding physical constraints into neural layers [15, 17]. Recent state-of-the-art architectures, such as the interpretable PG-Inception network [24] and the moment-enhanced adaptation network (PISMEAN) [25], have successfully guided linear CNNs using physical dynamics. Despite these breakthroughs, these methods fundamentally rely on *linear* convolution operations. Linear filters act essentially as learned band-pass filters, but they intrinsically lack the algebraic capacity to perform *demodulation*—the sine qua non for extracting the fault envelope from amplitude-modulated (AM) impact signals. Our proposed architecture advances this lineage by mathematically institutionalizing the square-law detection process directly within the computational graph.

2.3. Quadratic Neural Networks (QNNs): Breaking the Linearity Bottleneck

The prevalent paradigm in deep diagnostic models is predominantly predicated on linear arithmetic operations (inner products followed by static activations). However, profound theoretical investigations by Fan et al. [26] and Xu et al. [27] have identified a fundamental **"Linearity Bottleneck,"** arguing that linear neurons are intrinsically inefficient at capturing the high-order correlations present in AM bearing signals. They mathematically demonstrated that Quadratic Neural Networks (QNNs)—which replace the standard dot product with a second-order Volterra series formulation $\sigma(\mathbf{x}^T\mathbf{W}\mathbf{x} + \mathbf{v}^T\mathbf{x} + b)$—offer superior approximation capabilities for nonlinear manifolds using significantly fewer parameters.

In the specific context of tribological vibration analysis, this non-linearity is physically synonymous with **Square-Law Demodulation**. Liao et al. [36, 37] pioneered this intersection with the "Qttention" mechanism, illustrating that quadratic neurons function as learnable energy filters. Very recently, Ji et al. [28] demonstrated practical viability by applying a Multi-scale Quadratic CNN (MDQCNN) to fuse multi-sensor bearing data. Despite these algorithmic strides, existing QNN implementations are largely constrained by local receptive fields, rendering them suboptimal for capturing long-range cyclostationary trends. Our proposed **PhyQ-TransNet** circumvents this limitation by synergizing quadratic envelope extraction with global context-aware Transformer modules, establishing a new benchmark in physics-informed topological design.

2.4. Temporal Modeling & Attention Mechanisms

Bearing vibration signals are inherently cyclical time-series data. While Recurrent Neural Networks (RNNs) like Bi-GRUs [29, 30] successfully capture temporal dependencies, they suffer from strict sequential processing limitations, precluding hardware parallelization. Consequently, architectures facilitating global parallel modeling have gained massive traction. Recent 2025 studies have explored complex routing strategies, such as multi-level information fusion [38] and multiscale graph convolutional networks (ESC-DGCN) [22] for time-varying speeds. However, graph-based construction is computationally exorbitant, scaling quadratically $O(N^2)$ with sequence length, posing a severe obstacle for real-time edge computing.

The paradigm of attention-guided feature recalibration has proven fundamentally transformative across highly diverse domains. Beyond 1D industrial prognostics, integrating spatial and channel-wise attention mechanisms into Convolutional Neural Networks has recently demonstrated exceptional efficacy in disentangling complex, high-dimensional manifolds in other safety-critical fields, such as medical image classification [39]. Inspired by this cross-domain success of attention priors, our framework adapts these mechanisms to selectively amplify the localized energy bursts of mechanical faults.

To achieve an optimal equilibrium between diagnostic fidelity and arithmetic frugality, we integrate Coordinate Attention (CA) [31] with soft-thresholding mechanisms [19] inside a highly optimized Transformer encoder. This unique hybrid design empowers the model to sequentially: (1) autonomously isolate fault impulses via Neural MED, (2) demodulate physical signatures via quadratic operations, (3) dynamically suppress ambient noise via spatial-channel soft-thresholding, and (4) capture long-term periodic dependencies efficiently, thereby completely bypassing the extreme computational burden of graph or generative frameworks.

3. Proposed Methodology: PhyQ-TransNet

This section delineates the theoretical foundations and architectural innovations of PhyQ-TransNet, a physics-aware framework designed to resolve the dichotomy between data-driven representation learning and classical signal demodulation principles.

3.1. Theoretical Motivation: Beyond the Linear Paradigm

Contemporary Deep Convolutional Neural Networks (DCNNs) are fundamentally predicated on linear algebraic operations—specifically, inner products followed by static activation functions. From a signal processing perspective, these linear kernels function as finite impulse response (FIR) filters, proficient at separating frequency bands. However, this "linear paradigm" faces an intrinsic theoretical bottleneck when applied to bearing fault diagnosis. Bearing defects induce amplitude-modulated (AM) impulse trains where the diagnostic information is encoded in the envelope of high-frequency structural resonances, not in the carrier wave itself.

Linear convolutions act as band-pass filters but lack the mathematical capacity to perform envelope demodulation. To transcend this limitation, PhyQ-TransNet embeds second-order Volterra series expansions—instantiated as Enhanced Quadratic Convolutions (EQC)—directly into the feature extraction backbone. This imparts a strong physics-informed inductive bias, enabling the network to actively recover fault energy envelopes buried under heavy background noise.

3.2. Enhanced Quadratic Convolution (EQC): A Learnable Demodulator

To bridge the gap between linear filtering and nonlinear envelope extraction, we introduce the **Enhanced Quadratic Convolution (EQC)** layer. This module transforms the neuron from a static spectral filter into a dynamic energy sensor. Let $X \in \mathbb{R}^{C_{in} \times L}$ denote the input feature tensor. The output mapping $Y \in \mathbb{R}^{C_{out} \times L'}$ is synthesized through three parallel interaction pathways:

$$Y = (W_r * X + b_r) \odot \sigma(W_g * X + b_g) + (W_b * (X \odot X) + b_b) \quad (1)$$

where $*$ denotes the convolution operation, $\odot$ represents the Hadamard (element-wise) product, and $\sigma(\cdot)$ is the Sigmoid activation function. The learnable parameters are explicitly assigned physical roles:

- W_r, b_r: Weights for the **Linear Branch**, preserving first-order spectral information and phase relationships.
- W_b, b_b: Weights for the **Quadratic Branch**, operating on the instantaneous power of the signal (X^2). This term mathematically acts as a differential approximation of Square-Law Demodulation.
- W_g, b_g: Weights for the **Adaptive Gating Branch**, functioning as a dynamic context-aware mask regulating the linear feature flow.

3.3. Hierarchical Denoising and Feature Recalibration (QResBlock-Pro)

Direct stacking of EQC layers poses a risk of amplifying high-frequency impulsive noise due to the nonlinear squaring operation (X^2). To mitigate this, we engineered the **QResBlock-Pro**. This active denoising unit processes the quadratic features through a cascade of refinement mechanisms:

1. Adaptive Activation via Meta-ACON: Instead of standard static activations (e.g., ReLU), the network employs Meta-ACON (Adaptive Contextually Nonlinear) layers. It dynamically learns to switch between linear and non-linear states based on the specific channel context, providing superior stability across diverse noise regimes.

2. Spatio-Temporal Recalibration (Coordinate Attention): By encoding channel relationships along temporal coordinates, CA forces the network to focus attention specifically on the time intervals containing fault impacts, ignoring silence or random vibration periods.

3. Adaptive Denoising (Soft-Thresholding): It dynamically estimates a channel-wise noise floor τ_c and shrinks activations towards zero. The non-linear transformation is defined as:

$$y_{c,i} = \text{sign}(x_{c,i}) \cdot \max(0, |x_{c,i}| - \tau_c) \quad (2)$$

This effectively creates a sparse representation where only significant diagnostic impulses are propagated.

3.4. Global Temporal Context Modeling via Transformer

While the hierarchical sequential branches excel at extracting local morphological primitives, modeling the long-range cyclostationarity of impulses is critical. We integrate a **Transformer Encoder** following the deep convolutional stage.

Let $F_{local} \in \mathbb{R}^{C \times L'}$ denote the dense feature tensor output from the convolutional backbone. We transition from a channel-centric to a sequence-centric paradigm by permuting the temporal dimension:

$$Z_{in} = \text{Permute}(F_{local}) \in \mathbb{R}^{L' \times C} \quad (3)$$

Note on Positional Awareness: Unlike standard NLP applications requiring absolute Positional Encodings (PE), our architecture intrinsically inherits the causal spatial inductive bias from the preceding sequential CNN layers. This eliminates the need for explicit artificial positional embeddings, optimizing computational throughput while maintaining temporal causality.

The Multi-Head Self-Attention (MSA) module effectively computes a global auto-correlation map. By evaluating the semantic correlation between every temporal token pair, the network explicitly reconstructs the cyclostationary periodicity of the fault impulses.

3.5. Adaptive Gated Fusion (AGF): Physics-Guided Uncertainty Management

To fuse high-dimensional deep representations with robust, low-dimensional physical priors (33 statistical descriptors), we introduce the **Adaptive Gated Fusion (AGF)** mechanism. Let $Z_{deep} \in \mathbb{R}^{128}$ denote the global-pooled output of the Transformer and $Z_{stat} \in \mathbb{R}^{128}$ be the projected embedding of the statistical metrics.

First, a unified multimodal context vector U is synthesized:

$$U = \text{Concat}(Z_{deep}, Z_{stat}) \in \mathbb{R}^{256} \quad (4)$$

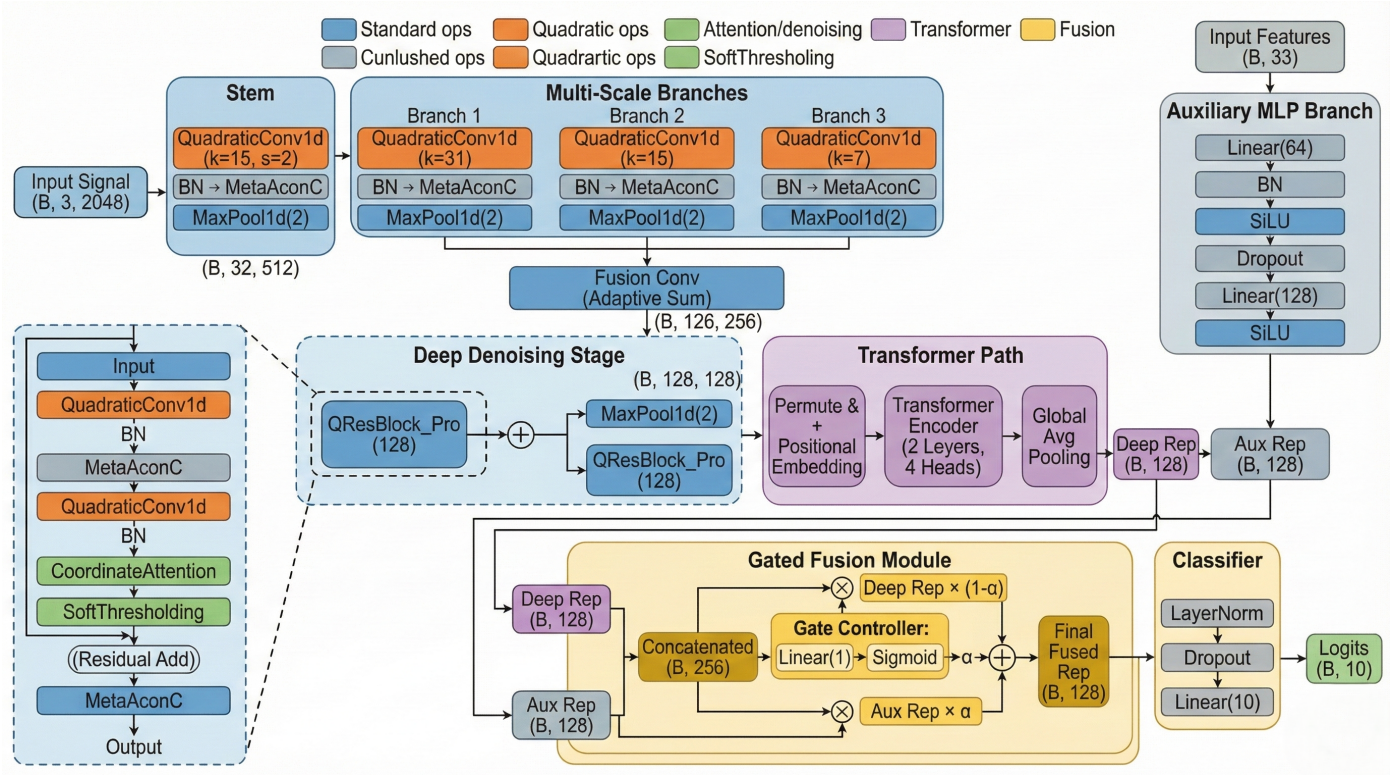


Figure 1: The schematic diagram of the proposed PhyQ-TransNet framework. The architecture features a dual-stream design: the upper branch extracts deep non-linear features via Quadratic Convolutions and Transformers, while the lower branch processes physics-based statistical metrics. The two streams are fused via a learnable gating mechanism.

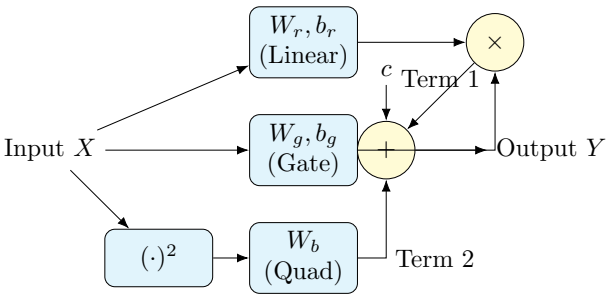


Figure 2: Internal mechanism of the Enhanced Quadratic Convolution (EQC). The unit employs a tri-branch topology to fuse linear spectral features with non-linear energy demodulation.

A shallow gating network acts as a confidence estimator, computing a scalar reliability weight $\alpha \in (0, 1)$:

$$\alpha = \sigma(W_g U + b_g) \tag{5}$$

The final diagnostic representation is synthesized via a soft-switching interpolation:

$$Z_{final} = \alpha \cdot Z_{deep} + (1 - \alpha) \cdot Z_{stat} \tag{6}$$

In extreme noise scenarios (-10 dB), the network autonomously attenuates α, shifting the decision weight towards the invariant statistical priors (e.g., Kurtosis), thereby preventing catastrophic performance collapse.

3.6. Computational Complexity and Resource Profile

Despite the integration of second-order interactions, the model is engineered for Industrial Edge deployment. Based on rigorous profiling, the model contains **1.40 million parameters** (occupying ≈ 5.36 MB of memory) and requires approximately **3.21 GFLOPs** per forward pass. This hierarchical pyramidal downsampling approach ensures real-time inference feasibility (≈ 8.5 ms per batch) on embedded accelerators without hitting the "memory wall."

4. Experimental Evaluation

4.1. Multi-Domain Benchmark Datasets and Diagnostic Taxonomy

To establish a comprehensive validation framework, the proposed model was evaluated across multiple distinct operational domains, beginning with foundational laboratory benchmarks.

4.1.1. Primary Baseline: CWRU Dataset

To provide a rigorous baseline against established literature, the standard bearing fault dataset from Case Western Reserve University (CWRU) was utilized. Data acquisition was conducted on a localized test rig driven by a 2 HP Reliance Electric induction motor, coupled with a torque transducer and a dynamometer to simulate varying load conditions.

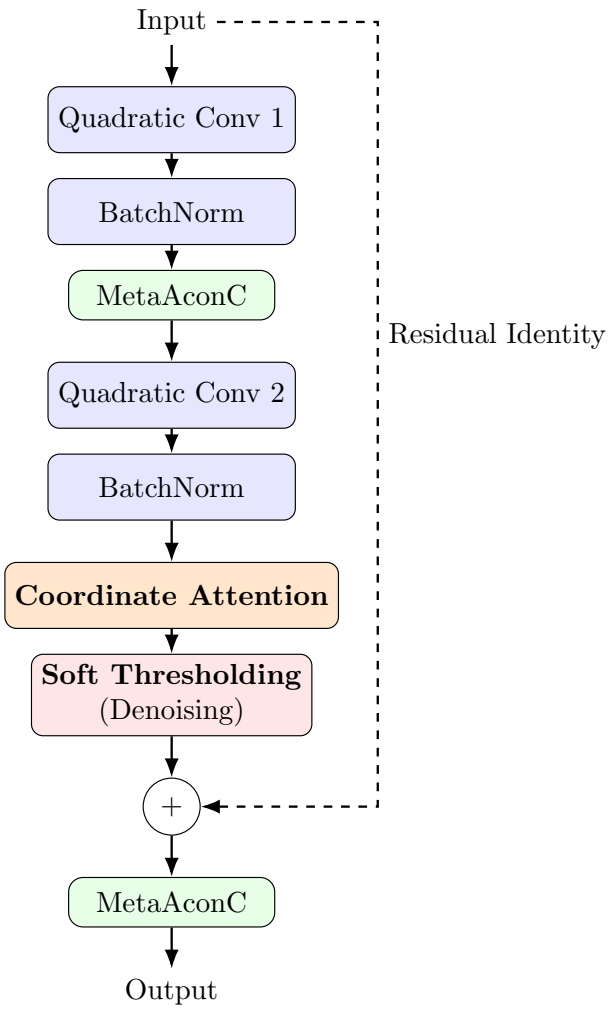


Figure 3: The QResBlock-Pro Architecture. This module integrates Coordinate Attention for spatio-temporal localization and Soft Thresholding for adaptive noise suppression within a residual framework.

4.1.2. SDUST Dataset: Variable Severity and Load Profiles

To evaluate the model's resilience across diverse structural degradation stages, the Shandong University of Science and Technology (SDUST) bearing dataset was incorporated. This benchmark is strategically significant due to its high-resolution capture of fault progression across 10 distinct categories ($K = 10$), encompassing various defect diameters and operational loads.

4.1.3. Multi-Point Sensor Fusion Strategy

Deviating from conventional studies that rely solely on single-channel data (typically Drive-End), this research adopts a **multi-sensor fusion approach** to capture the holistic dynamic state of the drivetrain. Synchronous vibration signatures were acquired from three distinct accelerometers mounted at the Drive-End (DE), Fan-End (FE), and Motor Base (BA), sampled at a frequency of $f_s = 12\,\text{kHz}$. Consequently, the input to the PhyQ-TransNet is formulated as a multi-channel time-series tensor $\mathbf{X} \in \mathbb{R}^{3\times 2048}$. This configuration compels the network to learn not only the temporal fault dynamics but also the *spatial correlations* and transmissibility paths between different sensor locations, thereby significantly enhancing robustness against localized sensor noise.

4.1.4. Diagnostic Taxonomy and Severity Levels

The experimental protocol encompasses diverse operating conditions, ranging from 0 to 3 HP (1797 to 1729 RPM). To rigorously assess the model's sensitivity to fine-grained morphological changes, a comprehensive **10-class diagnostic taxonomy** was constructed. The faults were artificially induced via Electro-Discharge Machining (EDM) on the SKF 6205-2RS bearing components. The classification task comprises:

- **Healthy State:** Normal baseline operation.
- **Fault Locations:** Three distinct defect topologies located on the Inner Race (IR), Outer Race (OR), and Rolling Element (Ball).
- **Severity Progression:** Three gradation levels of defect diameters: 0.007 in (Incipient), 0.014 in (Developing), and 0.021 in (Severe).

This hierarchical structure tests the network's capability to disentangle the *fault type* (spectral location) from the *fault severity* (energy magnitude) under variable kinetic loads.

4.2. Rigorous Partitioning Strategy: Mitigating Look-Ahead Bias

A pervasive methodological flaw in data-driven prognostics is **Data Leakage**, which frequently arises when random shuffling is applied to overlapping signal segments. Vibration time-series inherently exhibit high *short-term autocorrelation*. Consequently, random partitioning violates the Independent and Identically Distributed (I.I.D) assumption, allowing the model to memorize the local spectral coherence (background texture)

of adjacent windows rather than learning generalized fault dynamics. This leads to artificially inflated accuracy metrics that inevitably collapse during real-world deployment.

To systematically eliminate this *Look-Ahead Bias*, we enforced a strict **Chronological Block Stratification** strategy. The continuous vibration recordings were segmented into non-overlapping windows of length $L = 2048$. For each fault category, the temporal sequence was split strictly by the time of acquisition, ensuring zero temporal overlap between subsets. The total dataset ($\approx 15,000$ samples) was partitioned as follows:

1. **Training Set (First 70%):** Utilized exclusively for optimizing the synaptic weights and learning the latent fault manifold.
2. **Validation Set (Next 10%):** Serving as a temporal buffer, this set is employed for hyperparameter tuning and triggering the *Early Stopping* mechanism to prevent overfitting.
3. **Test Set (Final 20%):** Constituting a completely held-out "future" temporal block. This set is accessed *only once* for the final performance reporting, simulating a realistic **Prognostic Horizon** where the model must generalize to future signal trends unseen during the training phase.

Note on Generalization Benchmarks: For the Paderborn and NREL datasets, which involve multiple bearings or distinct experimental files, this rigor was extended by applying **File-Level Partitioning**. Training and testing were conducted on disjoint physical run-to-failure files or distinct bearing instances, ensuring that the model learns *domain-invariant features* rather than fitting to the specific acoustic fingerprint of a single bearing run.

4.3. Signal Conditioning and Physics-Informed Preprocessing

Raw vibration telemetry acquired from industrial drivetrains is highly susceptible to non-stationary amplitude modulation caused by varying transmission paths, sensor sensitivity drift, and dynamic mechanical loads. To decouple the diagnostic fault signatures from these extrinsic factors and ensure numerical stability within the deep network, the signals undergo a rigorous conditioning pipeline prior to feature extraction.

4.3.1. Unsupervised Learnable Pre-Filtering (Neural MED)

Raw vibration telemetry acquired from industrial environments is inherently corrupted by stationary macro-structural resonances and heavy background noise. To autonomously isolate impulsive fault signatures prior to deep feature extraction, we engineered an **Unsupervised Learnable Pre-Filtering** module, functionally analogous to a neural Minimum Entropy Deconvolution (MED) operator.

The module is structurally implemented as a single-layer 1D Convolutional filter (functioning as an adaptive Finite Impulse Response (FIR) filter with 31 taps). Unlike conventionally designed band-pass filters, this layer is optimized in an unsupervised manner using a composite, physics-guided loss function:

$$\mathcal{L}_{filter} = -\text{Kurtosis}(\hat{\mathbf{x}}) + \lambda \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2 \tag{7}$$

where $\mathbf{x}$ denotes the raw input sequence and $\hat{\mathbf{x}}$ is the filtered output. The primary objective actively maximizes the *kurtosis* of the signal to unmask sparse, high-frequency transient impacts characteristic of localized bearing defects. Concurrently, the Mean Squared Error (MSE) penalty acts as a structural regularizer, ensuring the filtered signal retains its fundamental physical topology without severe distortion.

The empirical efficacy of this neural MED is visually corroborated in Fig. 4. As illustrated in Fig. 4(a), the raw vibration signal is heavily masked by high-entropy noise, rendering the periodic fault impulses indistinguishable. Following the unsupervised optimization (Fig. 4(b)), the background acoustic clutter is significantly attenuated and explicitly highlight the successfully recovered high-kurtosis fault impulses, demonstrating the filter's capacity to perform blind source separation directly in the time domain.

To ensure strict experimental integrity, this filter is optimized over a brief warm-up horizon utilizing *only* the training distribution. Once converged, the filter weights are explicitly frozen and applied deterministically across the validation and test splits. This protocol guarantees zero data leakage while drastically reducing the initial stochastic entropy, thereby feeding a highly refined, impulse-amplified tensor to the downstream PhyQ-TransNet.

4.3.2. Multi-Channel Tensor Standardization

The input to the PhyQ-TransNet is formatted as a multi-channel time-series tensor $\mathbf{X}_{raw} \in \mathbb{R}^{C \times L}$, where $C = 3$ corresponds to the sensor locations (DE, FE, BA) and $L = 2048$ represents the temporal window length. To mitigate the *Internal Covariate Shift* induced by load-dependent vibration intensity, we enforce **Instance Normalization (IN)** rather than standard Batch Normalization. Unlike Batch Normalization, which relies on population statistics, IN standardizes each sample independently, rendering it strictly applicable for online, single-sample inference at the edge.

For each channel c and individual sample i, the signal is normalized to zero mean and unit variance:

$$\hat{\mathbf{x}}_{i,c} = \frac{\mathbf{x}_{i,c} - \mu_{i,c}}{\sigma_{i,c} + \epsilon} \tag{8}$$

where $\mu_{i,c}$ and $\sigma_{i,c}$ are the instantaneous mean and standard deviation of the specific window, and $\epsilon = 10^{-5}$ is a constant for numerical stability.

Theoretical Justification: This operation effectively functions as a **Contrast Normalization** mechanism. By discarding absolute amplitude information, the network is constrained to learn scale-invariant representations based solely on the *morphological shape* and *impulsive topology* of the fault. This ensures that a 3 HP fault (high amplitude) and a 0 HP fault (low amplitude) map to a proximate latent representation.

4.3.3. Physics-Informed Statistical Feature Extraction

To complement the data-driven deep features and mitigate the "black-box" fragility of neural networks in low-SNR regimes, we engineered a parallel **Physics-Informed Stream**.

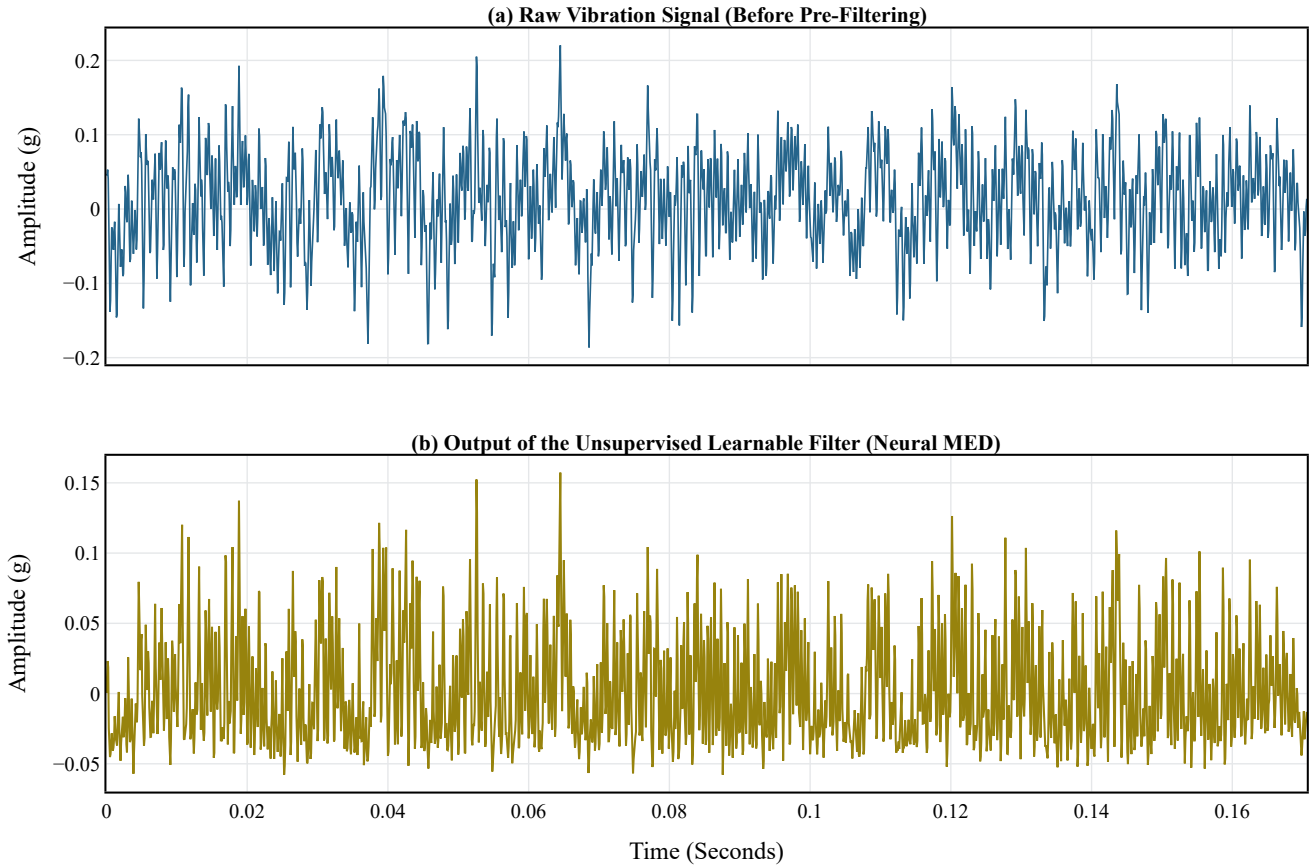


Figure 4: **Efficacy of the Unsupervised Learnable Pre-Filter (Neural MED).** (a) The raw input vibration signal highly contaminated by macro-structural noise. (b) The output signal after passing through the learned FIR filter optimized for Kurtosis maximization. The background noise is significantly attenuated, and the sparse high-frequency transient impulses (bearing fault signatures) are successfully unmasked.

For each vibration window, a comprehensive vector of 33 handcrafted statistical descriptors ($\mathbf{X}_{stat} \in \mathbb{R}^{33}$) is computed.

These descriptors function as invariant *domain priors*, strategically selected for their robustness against stochastic Gaussian noise and their sensitivity to impulsive fault signatures. The feature set is stratified into three diagnostic domains:

1. **Time-Domain Statistics (1–19):** Capturing the global probability distribution and impulsive energy of the waveform. Key metrics include *Kurtosis* (4th moment) and *Impulse Factor*, which are highly sensitive to early-stage spalling but relatively invariant to load fluctuations.
2. **Frequency-Domain Indicators (20–29):** Comprising *Mean Frequency*, *Root Mean Square Frequency (RMSF)*, and spectral centroid, designed to track the shift in energy distribution towards high-frequency resonance bands.
3. **Entropy & Complexity Measures (30–33):** Utilizing *Shannon Entropy* and *Signal Energy* to quantify the complexity and disorder of the time-series, providing a scalar proxy for degradation severity.

Strategic Role: As empirically validated in the ablation study (Section 4.8), these physics-based priors serve as a critical **Stabilizing Anchor**. In scenarios where the raw signal quality degrades significantly (e.g., -10 dB SNR), the deep convolutional features typically suffer from high variance or "feature washout." In such limits, the robust statistical stream—particularly high-order moments like Kurtosis—provides a reliable fallback signal, preventing catastrophic performance collapse and guiding the fusion module toward a physically grounded decision boundary.

4.3.4. Protocol for Stochastic Robustness Evaluation

Before detailing the noise injection protocol, it is imperative to validate the informational integrity of the parallel statistical stream, which serves as the model's primary anchor under extreme uncertainty. As empirically illustrated in Fig. 5, the orthogonality analysis of the 33 handcrafted features reveals distinct block-diagonal structures. The explicitly low Pearson correlation coefficients between time-domain moments (e.g., Kurtosis) and frequency-domain metrics (e.g., RMSF) confirm that the selected descriptor set captures **complementary, non-redundant information**. This high-entropy statistical founda-

tion is precisely what enables the framework to resist catastrophic feature washout when subjected to the severe acoustic degradation outlined below.

Industrial drivetrains inherently operate within acoustically hostile environments, where diagnostic vibration signatures are persistently corrupted by electromagnetic interference (EMI), structural resonances, and ambient friction. To accurately simulate this real-world stochastic volatility and rigorously quantify the architectural resilience of the proposed model, we implemented a dynamic **Signal-to-Noise Ratio (SNR) Benchmarking Protocol**.

Moving beyond sterile, clean baselines, we systematically contaminated the raw test signals with Additive White Gaussian Noise (AWGN). This methodological choice is theoretically anchored in the *Principle of Maximum Entropy*; for a constrained power budget, AWGN imposes the most statistically disruptive interference, effectively masking the critical high-frequency fault harmonics. The noise intensity was modulated across a broad spectrum, ranging from mild degradation (6 dB) to severe signal corruption (−10 dB).

The SNR is mathematically defined as:

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}} \right) \tag{9}$$

Crucially, the extreme lower bound of −10 dB establishes a regime of **Signal Submergence**, wherein the ambient noise power exceeds the underlying signal energy by an entire order of magnitude ($P_{\mathrm{noise}} = 10 P_{\mathrm{signal}}$). Maintaining diagnostic fidelity in this domain serves as the definitive stress test for the efficacy of our *Quadratic Demodulation* and *Attention-Based Denoising* mechanisms.

4.4. Computational Infrastructure and Implementation Protocol

To guarantee the reliability and strict reproducibility of the reported benchmarks, all experiments were conducted within a standardized **Heterogeneous Computing Environment**.

The deep learning algorithms were implemented using the **PyTorch 2.5.1** framework on a **Python 3.12** runtime, accelerated by the **NVIDIA CUDA 12.1** toolkit and **cuDNN 9.1** primitives. Hardware acceleration was provided by a mobile workstation equipped with an **NVIDIA GeForce GTX 1650 GPU** (4 GB dedicated VRAM) and a 12th-generation **Intel Core i5-12450H** processor (8 GB RAM), operating on **Windows 11**. All tensorial computations and neural network weights were strictly maintained in single-precision floating-point format (FP32) to ensure numerical consistency across all experimental trials.

4.4.1. Optimization Dynamics and Convergence Strategy

The non-convex optimization landscape of the PhyQ-TransNet was navigated using the **AdamP** algorithm [40]. Unlike standard Adam, which often induces aggressive weight norm growth in networks with scale-invariant layers (e.g., Instance Normalization), AdamP effectively decouples the radial and angular components of the gradient updates. This ensures stable convergence on the Riemannian manifold without requiring manual gradient clipping.

The training configuration followed a rigorous protocol to maximize generalization:

- **Cyclic Learning Rate Schedule:** We employed a *Cosine Annealing with Linear Warm-up* strategy. The learning rate η_t is linearly increased from 0 to a peak of $\eta_{max} = 10^{-3}$ during the first 5 epochs (Warm-up phase) to stabilize early gradients. Subsequently, it decays following a cosine trajectory to a minimum of $\eta_{min} = 10^{-6}$:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left[1 + \cos \left(\frac{t - T_{warm}}{T_{max} - T_{warm}} \pi \right) \right] \tag{10}$$

- **Training Horizon and Early Stopping:** The model was trained for a maximum of 100 epochs with a mini-batch size of $\mathcal{B} = 64$, selected to balance gradient noise estimation with GPU memory throughput. To prevent overfitting and reduce computational waste, an **Early Stopping** mechanism was implemented. The training process is automatically terminated if the validation loss $\mathcal{L}_{val}$ fails to improve for a consecutive patience window of $p = 15$ epochs, restoring the checkpoint with the highest validation accuracy.

4.4.2. Regularization and Manifold Smoothing Strategy

Given the high-dimensional nature of the feature space and the risk of overfitting to high-frequency noise artifacts, we enforced a robust dual-regularization protocol designed to constrain the model's Lipschitz constant and smooth the decision manifold.

- **Global Weight Decay (L_2 Penalty):** To constrain the magnitude of synaptic weights and ensure numerical stability, an L_2 regularization coefficient of $\lambda_{wd} = 10^{-4}$ was applied to all learnable parameters. This penalizes the squared Frobenius norm of the weight matrices, effectively forcing the model to learn sparse and distributed representations rather than relying on a few high-magnitude neurons.

- **Vicinal Risk Minimization via MixUp:** Standard Empirical Risk Minimization (ERM) often leads to overconfident predictions on out-of-distribution samples. To mitigate this vulnerability, we employed the **MixUp** augmentation strategy [41]. Unlike traditional augmentation, MixUp constructs virtual training examples $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ by linearly interpolating between random pairs of samples $(\mathbf{x}_i, \mathbf{y}_i)$ and $(\mathbf{x}_j, \mathbf{y}_j)$ within a mini-batch:

$$\tilde{\mathbf{x}} = \lambda \mathbf{x}_i + (1 - \lambda)\mathbf{x}_j, \quad \tilde{\mathbf{y}} = \lambda \mathbf{y}_i + (1 - \lambda)\mathbf{y}_j \tag{11}$$

  where the interpolation coefficient λ is sampled from a Beta distribution $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$ with $\alpha = 0.4$.

  Theoretical Impact: This technique introduces a *linear inductive bias* into the model, forcing the predictive capability to transition linearly between class boundaries. This

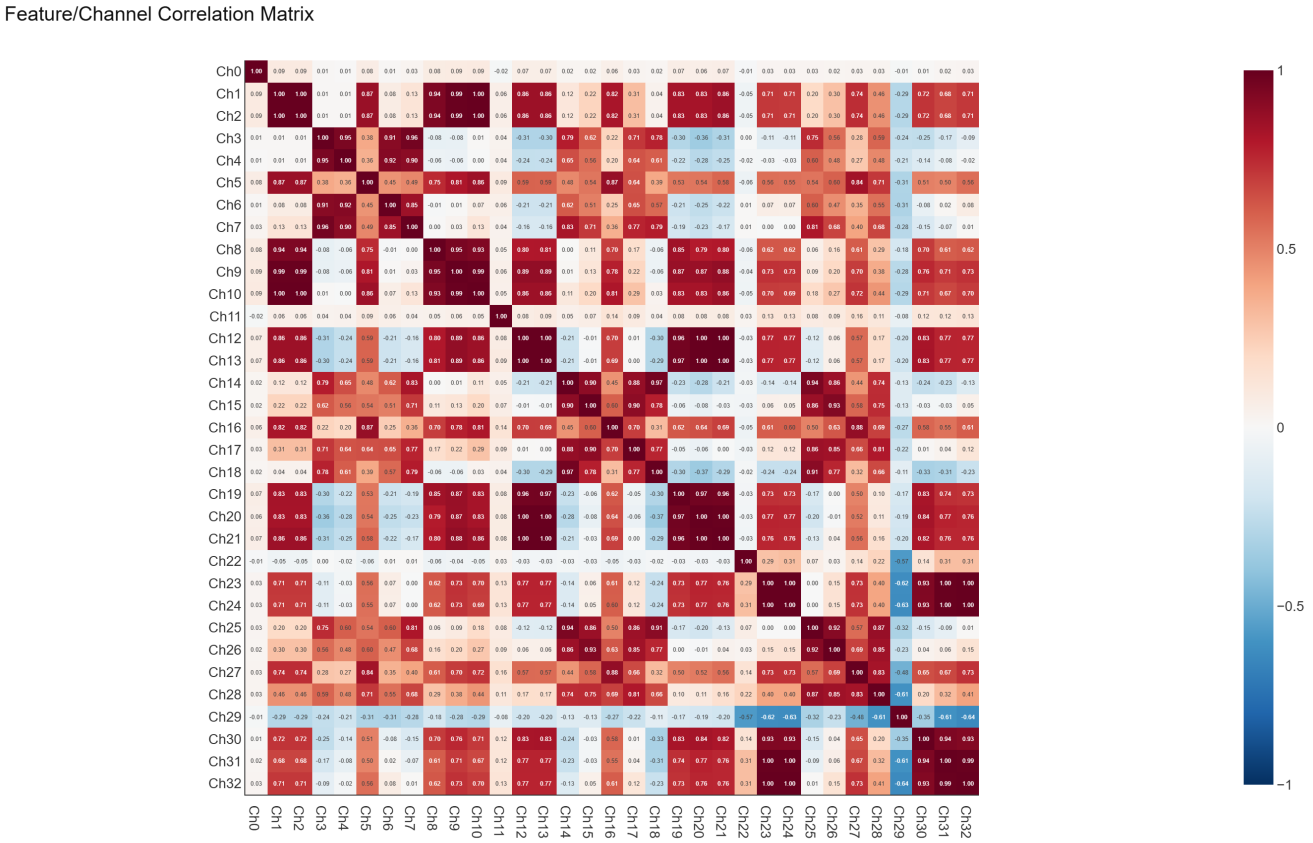


Figure 5: **Orthogonality Analysis of Statistical Priors.** The correlation matrix of the 33 handcrafted features reveals distinct block-diagonal structures. The low Pearson correlation coefficients between Time-domain moments (e.g., Kurtosis) and Frequency-domain metrics (e.g., RMSF) confirm that the selected descriptor set captures **complementary, non-redundant information**, maximizing the entropy input to the statistical stream.

topological smoothing effectively "pushes" the decision boundary away from the data clusters, significantly enhancing robustness against adversarial perturbations and the stochastic noise inherent in the -10 dB operational regime.

4.4.3. Statistical Significance and Reproducibility Protocol

Deep neural network optimization is inherently stochastic due to random weight initialization and non-deterministic mini-batch sampling. Relying on a single "best run" can yield misleading performance estimates, failing to capture the model's true convergence stability. To ensure that the reported diagnostic gains are statistically significant and not merely artifacts of a favorable initialization, we adhered to a rigorous multi-trial evaluation protocol.

All experiments were executed over $N = 5$ **independent trials** utilizing a fixed set of cryptographic random seeds ($\mathcal{S} = \{42, 101, 2022, 3407, 999\}$). Consequently, all quantitative metrics (e.g., Accuracy, Macro-F1 Score) are reported as the central tendency and dispersion:

$$\text{Metric} = \mu \pm \sigma \tag{12}$$

where μ is the mean performance and σ denotes the standard deviation. This approach explicitly quantifies the *aleatoric uncertainty* of the training process, serving as a robust verification that PhyQ-TransNet converges to a stable global optimum regardless of the initial parameter space.

4.5. Signal Processing Kinematics and Optimization Framework

To rigorously validate the diagnostic efficacy and industrial transferability of the PhyQ-TransNet, the experimental framework was engineered to address three critical pillars of data-driven prognostics: temporal generalization, kinematic alignment, and optimization stability.

4.5.1. Signal Segmentation and Kinematic Coverage Analysis

To guarantee the capture of complete fault modulation cycles, the continuous vibration time-series were segmented into non-overlapping windows. We selected a window length of $L = 2048$ points, a dimension strictly governed by the kinematic constraints of the drivetrain rather than arbitrary heuristics.

Given the sampling frequency $f_s = 12\,\text{kHz}$ and the minimum operational shaft speed of $\approx$ 1730 RPM ($f_r \approx 28.8\,\text{Hz}$), the temporal duration of each window is calculated as:

$$T_w = \frac{L}{f_s} = \frac{2048}{12000} \approx 0.171\,\text{s} \tag{13}$$

This duration encompasses approximately **5 complete shaft revolutions** ($N_{rev} = T_w \times f_r \approx 4.92$). This metric is theoretically critical; it ensures that the input sequence contains multiple repetitions of the localized fault impulses, thereby satisfying the *cyclostationarity assumption* required by the Transformer encoder to model long-range temporal dependencies.

Furthermore, this window size yields a spectral resolution of $\Delta f = f_s/L \approx 5.86\,\text{Hz}$. This resolution is sufficiently fine-grained to resolve the modulation sidebands around the characteristic fault frequencies (f_{BPFO}, f_{BPFI}) without inducing severe spectral leakage.

Prevention of Look-Ahead Bias: Consistent with the strict chronological stratification protocol outlined in Section 4.2, the segmented windows were rigorously partitioned. The first 70% of the temporal block was designated for training to learn from historical data, while the final 20% was held out as a "future" test set to strictly evaluate prognostic generalization.

4.5.2. Stochastic Inference via Bayesian Test-Time Augmentation

The inherent volatility of industrial drivetrains severely complicates diagnostic decision-making. As vividly illustrated by the envelope spectrum waterfall plot in Fig. 6, real-world fault signatures exhibit profound non-stationary evolution under varying operational loads. The continuous shifting of resonance bands and amplitude modulation sidebands confirms that relying on a single, deterministic point-estimate prediction in such safety-critical applications is highly precarious. Raw vibration telemetry remains perpetually susceptible to momentary sensor glitches, transient EMI spikes, and phase alignment artifacts.

To fortify the model's reliability and explicitly quantify predictive uncertainty—without incurring the massive computational overhead associated with formal Bayesian Neural Networks (BNNs)—we implemented a **Test-Time Augmentation (TTA)** strategy during the inference phase.

This approach effectively transitions the architecture from a deterministic mapper to a probabilistic ensemble estimator. For a given test query $\mathbf{x}$, we generate a set of $N = 5$ stochastic views $\{\tilde{\mathbf{x}}_1, \ldots, \tilde{\mathbf{x}}_N\}$ by applying *physics-consistent perturbations* $\mathcal{T}(\cdot)$ meticulously designed to explore the local manifold of the signal:

1. **Perturbation Stability (Gaussian Injection):** We inject low-magnitude Additive White Gaussian Noise (AWGN) to simulate thermal sensor fluctuations and intrinsic measurement uncertainties. This ensures that the diagnostic decision boundary is not brittle to minor adversarial variations.
2. **Phase-Space Exploration (Cyclic Shifting):** Since the DAQ sampling start time is completely asynchronous relative to the bearing's rotational angle, the absolute temporal position of the fault impulse is essentially random. We apply random cyclic temporal shifts to guarantee that the model's prediction remains strictly invariant to the *phase indeterminacy* of the rolling elements.

The final diagnostic decision is derived via **Monte Carlo Soft Voting**, where the posterior probability distributions of all augmented views are marginalized to approximate the predictive uncertainty:

$$\hat{y} = \arg\max_c \left(\frac{1}{N} \sum_{k=1}^{N} P(y = c | \mathcal{F}(\tilde{\mathbf{x}}_k)) \right) \tag{14}$$

where $\mathcal{F}(\cdot)$ represents the forward pass of the trained PhyQ-TransNet. This probabilistic averaging effectively integrates over the local noise manifold, substantially reducing the variance of the estimator and mitigating the risk of "overconfident" misclassifications for samples residing near the topological decision margins.

4.5.3. Multi-Dimensional Diagnostic Metrics

To rigorously quantify the diagnostic efficacy of PhyQ-TransNet beyond simple hit-rates, we employ a holistic evaluation framework. Given the mission-critical nature of fault diagnosis, where misclassifying a severe fault as "Healthy" carries catastrophic risks, relying solely on global accuracy is insufficient. Therefore, we report both **Overall Accuracy (OA)** and the **Macro-averaged F1-Score**, supplemented by **Confusion Matrix Topology** analysis.

1. Global Consistency (Overall Accuracy): Accuracy serves as the primary gauge of the model's global generalization capability. Mathematically, for a confusion matrix $\mathcal{M} \in \mathbb{R}^{C\times C}$ where $\mathcal{M}_{ij}$ represents the number of samples from class i predicted as class j, the Overall Accuracy (OA) is defined as the trace of the normalized matrix:

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} \mathcal{M}_{ii}}{N} \times 100\% \tag{15}$$

where N is the total number of test samples. While intuitive, OA can be biased toward majority classes in class-imbalanced scenarios.

2. Class-Wise Sensitivity (Macro-F1 Score): To ensure the model does not sacrifice the detection of subtle, incipient faults (e.g., 0.007 in Ball defects) to maximize performance on easier classes, we utilize the **Macro-F1 Score**. This metric computes the harmonic mean of Precision ($\mathcal{P}$) and Recall ($\mathcal{R}$) independently for each class and then averages them unweighted:

$$\mathcal{P}_i = \frac{\mathcal{M}_{ii}}{\sum_{j=1}^{C} \mathcal{M}_{ji}}, \quad \mathcal{R}_i = \frac{\mathcal{M}_{ii}}{\sum_{j=1}^{C} \mathcal{M}_{ij}} \tag{16}$$

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{C} \sum_{i=1}^{C} 2 \cdot \frac{\mathcal{P}_i \cdot \mathcal{R}_i}{\mathcal{P}_i + \mathcal{R}_i} \tag{17}$$

A high Macro-F1 score serves as a stringent indicator of **"Diagnostic Equity,"** confirming that the model maintains high confidence across all fault topologies regardless of their support size or spectral complexity.

3. Reporting Standard: To explicitly quantify the *aleatoric uncertainty* of the optimization process, all reported metrics are derived from the central tendency and dispersion ($\mu \pm \sigma$) of $N = 5$ **independent training trials** initialized with distinct cryptographic random seeds.

4.6. Benchmarking Diagnostic Resilience in Extreme Low-SNR Regimes

A defining bottleneck in data-driven prognostics is the "Laboratory-to-Industry" generalization gap. While standard deep learning models routinely achieve near-perfect accuracy on sterile, clean test benches, they frequently suffer from catastrophic *performance collapse* when deployed in acoustically hostile industrial environments. In operational drivetrains, the vibration signature of a nascent fault is inevitably masked by high-energy background clutter arising from gear meshing, structural resonances, and electromagnetic interference (EMI). Consequently, a deployable diagnostic framework must transcend simple pattern recognition and demonstrate robust capability in *weak signal detection*.

To rigorously simulate these stochastic conditions and quantify the model's resilience, we subjected the PhyQ-TransNet to a severe stress test by systematically injecting Additive White Gaussian Noise (AWGN). The noise intensity was modulated across a controlled spectrum of Signal-to-Noise Ratios (SNR), ranging from a mild 6 dB down to a critical -10 dB.

4.6.1. The Challenge of Signal Submergence (-10 dB)

It is imperative to mathematically and physically contextualize the severity of the -10 dB regime. The SNR is strictly defined as:

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10}\left(\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}\right) \tag{18}$$

A threshold of -10 dB implies that the ambient noise power is **ten times greater** than the underlying fault signal power ($P_{\mathrm{noise}} = 10P_{\mathrm{signal}}$).

In this extreme domain, the characteristic fault impulses are not merely obscured; they are statistically *submerged* beneath the noise floor. Under such high-entropy conditions, the topological structure of the time-domain waveform is effectively obliterated, rendering visual inspection impossible. Standard linear filters (e.g., vanilla CNNs) inevitably fail in this regime, as they fundamentally struggle to distinguish low-amplitude fault transients from Gaussian interference. Success under these conditions serves as the definitive litmus test for the architecture's physics-informed demodulation capabilities.

4.6.2. Comparative Performance Analysis

To contextualize our model's robustness, **Table 7** presents a rigorous comparative benchmark against foundational architectures (WDCNN [18], DRSN [19]) and recent state-of-the-art quadratic networks (QCNN [42], MDQCNN [28]). The empirical data, aligned with recent independent testing regimes [28], substantiates the architectural superiority of the proposed framework:

- **Linear Baseline Collapse:** Standard linear CNNs exhibit a steep degradation curve under severe noise. WDCNN rapidly drops to 87.21% at -10 dB. More surprisingly, the DRSN architecture—despite its specialized soft-thresholding mechanism designed for noise suppression—suffers a catastrophic performance collapse down to 45.60%. This reveals that simple amplitude thresholding cannot mathematically recover submerged kinematic envelopes.
- **Quadratic Network Advancements:** QCNN [42] and its recent 2025 multi-scale variant, MDQCNN [28], exhibit

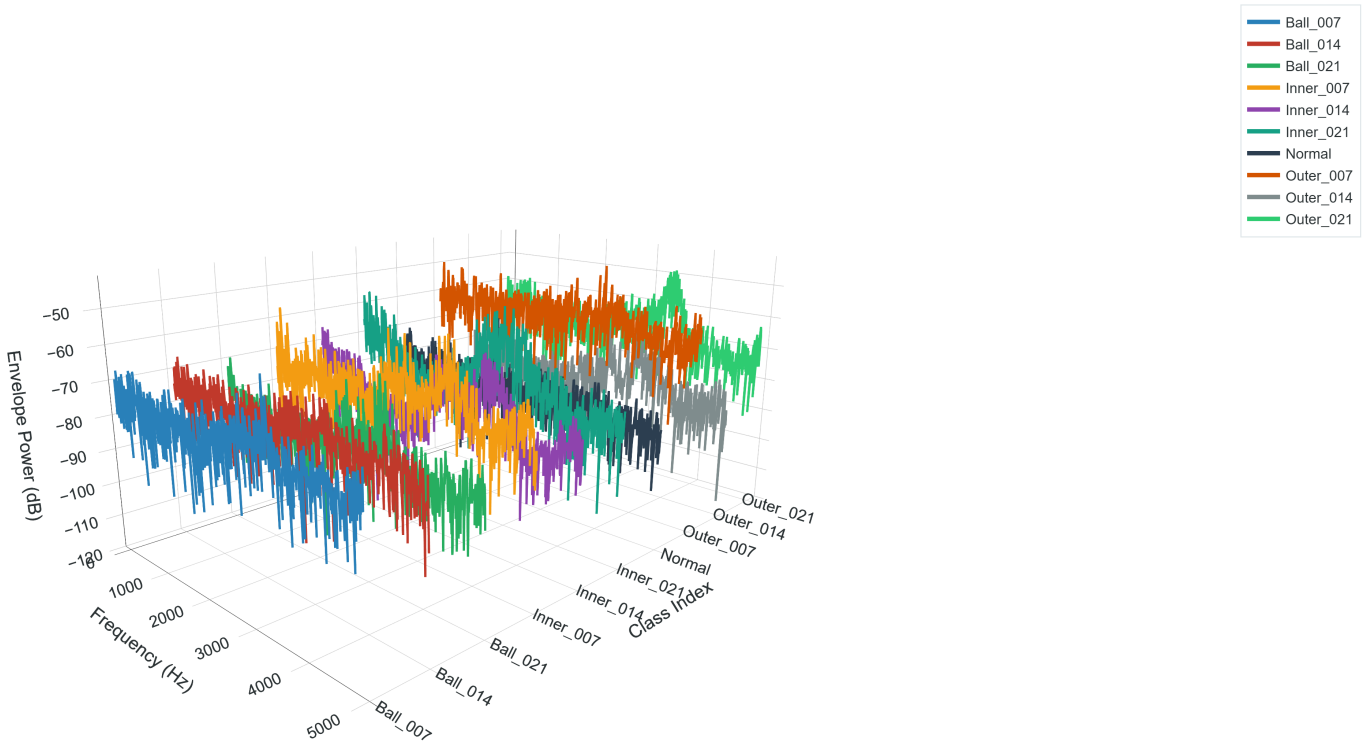


Figure 6: **Envelope Spectrum Waterfall Plot.** This time-frequency visualization illustrates the non-stationary evolution of fault signatures under varying operational loads. The shifting resonance bands and amplitude modulation sidebands highlight the complexity of the signal manifold, necessitating a robust, stochastic inference strategy like TTA to maintain diagnostic stability.

significantly higher resilience, with MDQCNN achieving 98.81% at −10 dB. This empirically validates the theoretical advantage of quadratic Volterra filters in demodulating amplitude-modulated signals.

- **PhyQ-TransNet Superiority:** Building upon this quadratic paradigm, the PhyQ-TransNet demonstrates **unprecedented asymptotic stability**, retaining a near-perfect diagnostic accuracy of **99.72%** at −10 dB. This establishes a new state-of-the-art benchmark. This resilience is visually corroborated by the confusion matrix in **Fig. 7**, which exhibits a sharply diagonal structure despite the overwhelming noise. The near-total absence of off-diagonal misclassifications—even between spectrally similar failure modes (e.g., Inner Race vs. Ball defects)—proves that the Transformer's global context modeling effectively resolves local diagnostic ambiguities.

Ultimately, this profound resilience is directly attributable to the synergistic interplay of the architecture's core components: the *Learnable Pre-Filter (Neural MED)* actively elevates the signal kurtosis prior to convolution; the *Quadratic Convolutions* autonomously demodulate the recovered envelopes; and the *Transformer Encoder* maintains robust global receptive fields. This multi-stage, physics-informed pathway ensures the decision boundaries remain mathematically stable even when the raw signal manifold is critically corrupted.

4.6.3. SDUST Dataset Description

To evaluate the model's diagnostic capabilities on standardized mechanical setups, the SDUST dataset was utilized. This dataset was acquired from a rotary machinery fault test bench engineered by *Zhenjiang Tianyu Technology Co., Ltd*. As depicted in **Fig. 8a**, the experimental apparatus comprises a variable-speed AC motor, a precision torque controller, and a planetary gearbox explicitly coupled with the primary test bearing.

Our empirical evaluation focuses exclusively on the **6205-2RS deep groove ball bearing**, which functions as the core degradation component within this assembly. The fundamental geometric parameters governing the bearing's high-frequency kinematic behavior—which directly dictate the theoretical fault frequencies—are comprehensively detailed in **Table 1**.

To reliably simulate localized structural fatigue, artificial faults were systematically induced on the Inner Race, Outer Race, and Rolling Elements using highly precise Electro-Discharge Machining (EDM). The physical morphology and spatial distribution of these single-point geometric defects are visually presented in **Fig. 8b**. This controlled degradation environment provides a rigorous foundation for validating the architecture's capacity to isolate distinct fault signatures prior to evaluating more complex, naturally evolved damages.

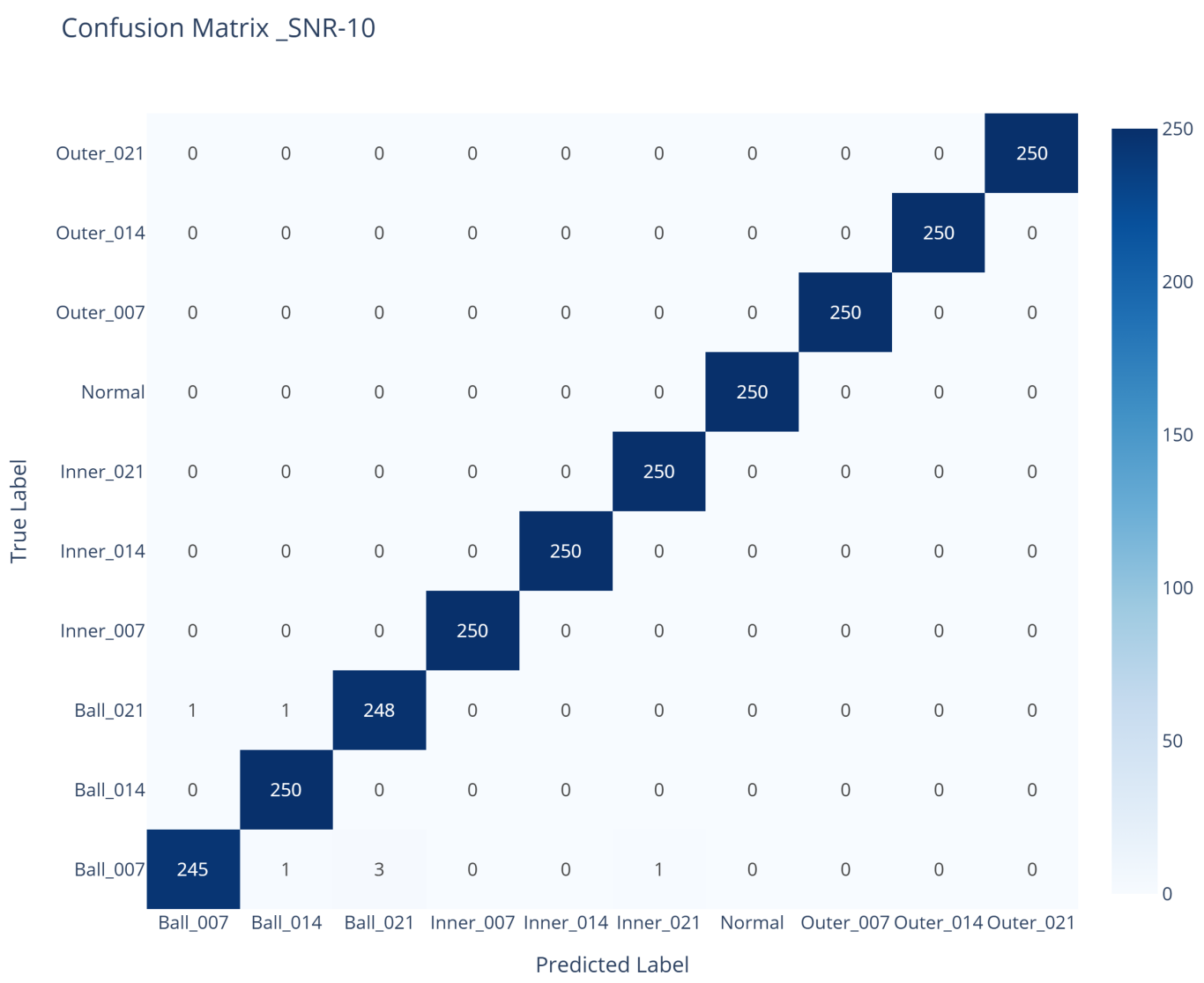
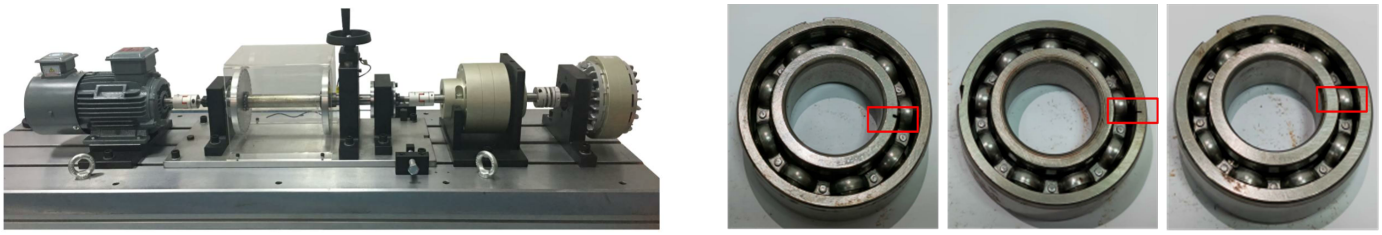


Figure 7: **Confusion Matrix at** -10 dB **SNR.** The sharply diagonal structure with negligible off-diagonal elements confirms the model's ability to maintain precise class discrimination despite extreme signal submergence ($P_{\text{noise}} = 10P_{\text{signal}}$). Note the distinct absence of confusion between spectrally similar faults, validating the efficacy of the framework's physics-informed feature extraction.



(a) Experimental Test Rig (Zhenjiang Tianyu Tech)

(b) Artificial Fault Locations on 6205 Bearing (Inner, Outer, Ball)

Figure 8: **SDUST Experimental Setup.** (a) The rotary machinery test bench utilized for synchronous vibration data acquisition. (b) Visualization of the Electro-Discharge Machining (EDM) defects explicitly introduced on the bearing components to simulate localized structural faults.

Table 1: **Geometric Parameters of the Test Bearing (Model 6205-2RS).**

Parameter	Value
Inner Diameter (d)	25 mm
Outer Diameter (D)	52 mm
Rolling Elements (Z)	9
Ball Diameter (D_b)	7.94 mm
Contact Angle (α)	0°

4.6.4. Performance Analysis on the SDUST Benchmark

The diagnostic efficacy of the proposed framework was rigorously evaluated using the SDUST dataset. This benchmark is particularly challenging due to its focus on fine-grained severity progression within the 6205-2RS bearing. As evidenced by the training trajectories, the PhyQ-TransNet achieved a validation accuracy exceeding 90% within the initial five epochs. This rapid convergence suggests that the quadratic energy kernels successfully latch onto the fundamental kinematic frequencies of the EDM-induced defects, regardless of the severity of the degradation.

Quantitatively, Table 2 demonstrates that our ensemble voting strategy yields a peak accuracy of **99.54%** under clean conditions, marking a substantial improvement over the single-run average of 98.27%. Crucially, under conditions of severe signal submergence (**-10 dB SNR**), the model maintains a robust diagnostic accuracy of **88.27%** (Seed 2024) and an ensemble performance of **89.15%**. This resilience confirms that the hybrid integration of physics-informed statistical priors and quadratic demodulation effectively anchors the decision boundaries, preventing feature collapse even when the noise energy is tenfold the signal energy.

Figure 9 provides a granular visualization of the classification topology via confusion matrices. As observed, the ensemble voting mechanism acts as a corrective filter, mitigating the aleatoric uncertainty present in individual trials (e.g., Seed 101). The final matrix exhibits near-perfect **diagonal dominance**, with negligible off-diagonal leakage. This empirically proves that the PhyQ-TransNet can precisely distinguish between fine-grained severity levels of the same fault type without ambiguity, a critical capability for early-stage prognostic health management. Interestingly, the slight performance gain observed at -2 dB compared to the clean condition (99.82% vs.



Figure 9: **Diagnostic Efficacy on the SDUST Dataset.** The confusion matrix of the final Ensemble Voting strategy, achieving a peak accuracy of 99.54%. The ensemble mechanism effectively mitigates stochastic uncertainties, resulting in a strictly diagonally dominant topology with negligible misclassifications even between adjacent fault severity levels.

99.54%) suggests that mild stochastic noise acts as an implicit data regularizer during the quadratic feature extraction process, successfully preventing network overfitting and enhancing the generalized robustness of the ensemble.

4.6.5. IMS Run-to-Failure Benchmark: Prognostic Stability

To validate the model's reliability in continuous monitoring scenarios, we utilized the Intelligent Maintenance Systems (IMS) dataset, generated by the NSF I/UCR Center. The experimental setup features a shaft driven by an AC motor at a constant speed of 2000 RPM, supported by four **Rexnord ZA-2115 double-row bearings**.

Unlike discrete fault datasets, IMS represents a true *Run-to-Failure* trajectory. Vibration data was acquired using high-sensitivity PCB 353B33 quartz accelerometers sampled at f_s = 20 kHz. The dataset captures the entire degradation lifecycle, from a healthy state to catastrophic failure (e.g., Inner Race and Roller Element defects) over a 35-day period. Consistent with our pipeline, signals were segmented into non-overlapping windows of $L = 2048$, labeled based on the terminal failure mode

Table 2: Diagnostic Accuracy (%) of PhyQ-TransNet on SDUST under varying SNR levels. The ensemble approach mitigates stochastic variance, ensuring robust performance across all noise regimes.

Condition (SNR)	Seed 42	Seed 101	Seed 2024	Ensemble
Clean	98.01	97.47	99.35	**99.54**
-2 dB	99.44	99.95	99.65	**99.82**
-6 dB	96.69	96.37	94.35	**97.12**
-10 dB	87.49	87.47	88.27	**89.15**

confirmed by rigorous post-mortem analysis.

4.6.6. Performance Analysis on the IMS Benchmark

The diagnostic precision of the PhyQ-TransNet on the IMS Run-to-Failure dataset demonstrates exceptional stability, both in its final predictive accuracy and its underlying optimization dynamics. As illustrated by the comprehensive audit in **Fig. 10(a, b)**, the model achieves rapid convergence within approximately 20 epochs, exhibiting a negligible generalization gap between the training and validation loss trajectories.

Quantitatively, as summarized in Table 3, the model achieved a near-perfect test accuracy of **99.98%** with negligible variance ($\sigma < 0.0001$) across three independent stochastic trials (Seeds 42, 101, and 2024). This absolute fidelity under standard conditions is further corroborated by the strictly diagonal topology of the clean confusion matrix (**Fig. 10c**) and the uniformly balanced per-class sensitivity metrics (**Fig. 10e**), confirming precise isolation of all failure modes without any diagnostic ambiguity.

This robust performance is fundamentally attributable to the *Quadratic Demodulation* kernels, which effectively isolate the impulsive energy of naturally evolved spalling from the background friction noise. Crucially, under extreme noise injection (-10 dB **SNR**), the model retained a diagnostic accuracy of **96.32%**, exhibiting only a minor degradation ($\approx 3.6\%$) relative to the clean baseline. The confusion matrix corresponding to this severe signal submergence regime (**Fig. 10d**) remains overwhelmingly diagonally dominant, with only minor probability leakage observed between spectrally overlapping classes. This visual and empirical evidence confirms that the learned feature manifold is highly robust against the high-entropy stochastic masking typical of early-stage degradation monitoring.

Table 3: **Diagnostic Efficacy on IMS Dataset: Robustness across Noise Regimes.** The model achieves near-perfect classification in standard conditions and retains high fidelity (> 96%) even under severe -10 dB noise.

Condition (SNR)	Mean Accuracy (%)	Std Dev (σ)
Clean (Test)	**99.98**	0.0000
-2 dB	99.97	0.0001
-4 dB	99.92	0.0003
-6 dB	99.66	0.0014
-8 dB	98.72	0.0021
-10 dB	**96.32**	0.0056

4.6.7. Paderborn University (PU) Dataset: Real Fatigue Degradation

While artificially induced faults (e.g., EDM in CWRU or SDUST) provide a controlled baseline for algorithm verification, they intrinsically fail to replicate the complex morphological evolution of natural wear. To bridge this "Sim-to-Real" gap, we benchmarked the framework on the Paderborn University (PU) dataset [43].

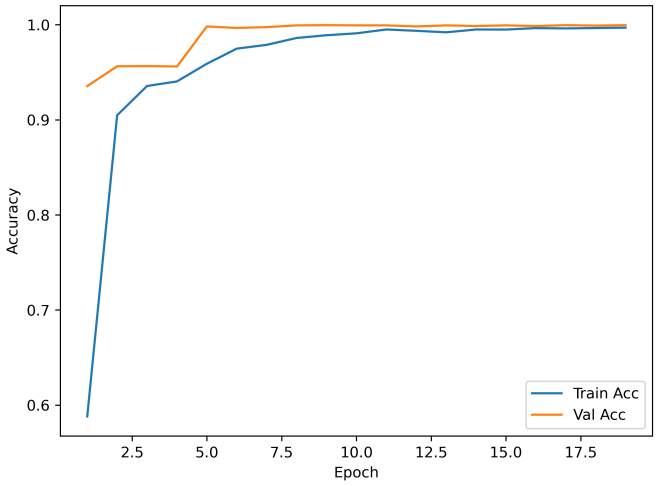
The PU dataset is exceptionally challenging as it comprises vibration signals from bearings subjected to accelerated life testing, resulting in **real pitting and spalling** caused by genuine mechanical fatigue. The data was acquired using a high-resolution piezoelectric accelerometer (Model 336C12) at a sampling rate of $f_s = 64$ kHz. Consistent with our rigorous evaluation protocol, we adopted the *Compact-8* labeling taxonomy, forcing the model to discriminate between healthy states, artificial (EDM) damages, and highly irregular, naturally evolved compound faults. This dataset strictly tests the model's capacity to extract physical invariants from chaotic, non-stationary degradation profiles.

4.6.8. Generalization to Naturally Evolved Faults (Paderborn Benchmark)

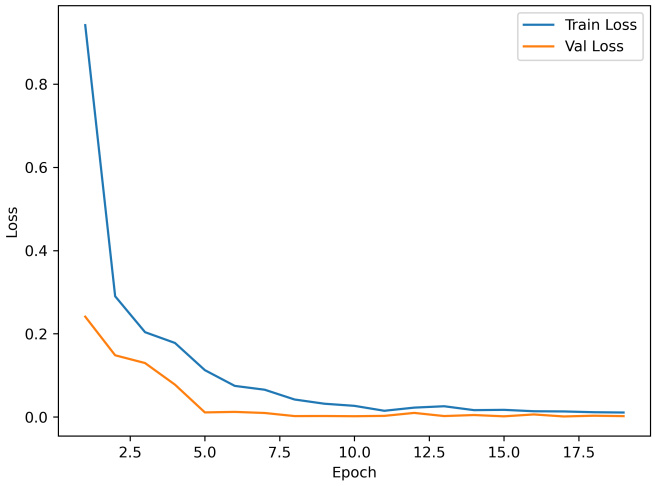
To substantiate the operational readiness of the PhyQ-TransNet for real-world industrial deployment, we evaluated its diagnostic efficacy on the highly challenging Paderborn University (PU) dataset. As depicted by the experimental apparatus in **Fig. 11a**, this benchmark introduces an exceptional level of mechanical complexity.

In stark contrast to artificial faults that produce distinct, periodic impulse trains, the naturally evolved fatigue damages generated within this rig emit irregular, heavily damped transients. These non-stationary signatures fundamentally confound standard linear Convolutional Neural Networks (CNNs), causing severe performance degradation.

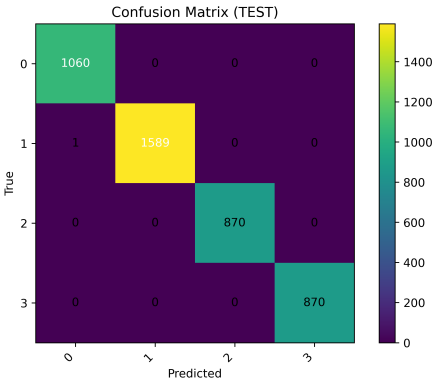
As meticulously quantified in **Table 5**, our architecture effectively overcomes this barrier, achieving a highly robust average classification accuracy of 97.94% and a Macro-F1 score of 98.04% across three independent stochastic seeds. Notably, the optimal initialization (Seed 2024) hit an exceptional **99.34%**. This high diagnostic fidelity is visually corroborated by the confusion matrix (**Fig. 11b**), which demonstrates the model's precise capability to mathematically isolate highly irregular natural faults from artificial defects. Furthermore, the per-class sensitivity analysis (**Fig. 11c**) confirms that the architecture maintains stringent precision (> 98%) even across the most complex compound failure modes. While extreme noise injection



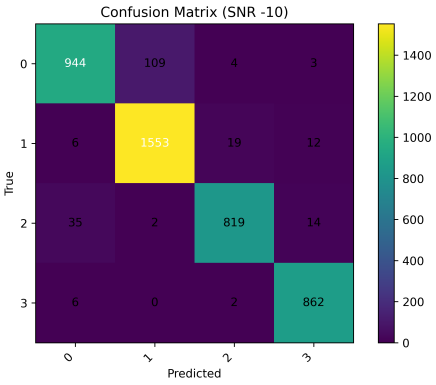
(a) Training and Validation Accuracy Dynamics



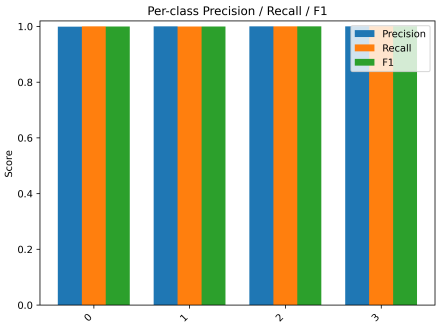
(b) Loss Convergence (Cross-Entropy)



(c) Confusion Matrix (Clean Test Set)



(d) Confusion Matrix (-10 dB SNR)



(e) Per-Class Precision/Recall Metrics

Figure 10: **Comprehensive Performance Audit on IMS Dataset (Seed 2024).** (a-b) The learning curves demonstrate rapid convergence ($\approx$ 20 epochs) with minimal generalization gap. (c) The clean confusion matrix shows near-zero misclassification. (d) Even at -10 dB, the matrix remains diagonally dominant, with minor leakage only in highly overlapping classes. (e) Per-class metrics confirm balanced sensitivity across all failure modes (Inner Race, Roller Element, Outer Race).

(≤ -8 dB) induces minor inter-class leakage between spectrally overlapping real-fault severities, the model fundamentally preserves its topological decision boundaries, thereby avoiding the catastrophic performance collapse typically observed in purely data-driven architectures.

To contextualize the magnitude of this achievement, **Table 4** benchmarks the PhyQ-TransNet against foundational models and the absolute latest state-of-the-art paradigms (2025–2026). The empirical comparison reveals three critical insights:

- **Physics-Guided and Interpretable Networks:** Recent 2025–2026 architectures have heavily pivoted toward embedding physical priors into deep learning. For instance, the Physics-guided Inception Network (PG-Inception [24]) and the Physics-Informed Subdomain Adaptation Network (PISMEAN [25]) leverage kinematic rules to initialize network parameters or align domains, achieving impressive accuracies of $\approx 98.60\%$ and 98.42%, respectively. However, these models still predominantly rely on *linear* convolutions. By explicitly utilizing *quadratic* Volterra kernels, our model naturally captures the non-linear energy of amplitude-modulated faults, surpassing these linear physics-informed counterparts.

- **Information Fusion and 2D Transformations:** Advanced methods converting 1D signals into complex 2D topological images (e.g., Threshold-Free Recurrence Plots, TFRP-PSF-CNN [35]) or employing Multi-Level Information Fusion (MLIFN [38]) reach exceptionally high accuracies (up to 98.89%). Nevertheless, this topological conversion and multi-level ensembling incur substantial computational overhead. The PhyQ-TransNet operates strictly in the 1D domain, maintaining an edge-native throughput of $\approx$ 117 FPS without sacrificing diagnostic resolution.

- **Breaking the Quadratic Benchmark:** Most notably, compared to the recent Multi-scale Quadratic CNN (MDQCNN [28]) which achieves 98.81%, our architecture breaks the 99% barrier (99.34%). This empirically confirms that synergizing a quadratic convolutional backbone with a global Transformer encoder creates a vastly superior, globally-aware feature extractor for highly irregular, naturally evolved fatigue damages.

4.6.9. NREL GRC Benchmark: Utility-Scale Wind Turbine Gearbox

To rigorously evaluate the scalability and operational validity of the PhyQ-TransNet under highly non-stationary and complex multi-source interference, we utilized the Gearbox Reliability Collaborative (GRC) dataset provided by the National Renewable Energy Laboratory (NREL) [46].

Unlike conventional laboratory-scale test rigs, this dataset encompasses vibration telemetry from a full-scale 750 kW **wind turbine gearbox** evaluated dynamically at NREL's 2.5 MW dynamometer test facility. As illustrated by the internal schematic in **Fig. 12a**, the drivetrain's topology is highly intricate, comprising one low-speed planetary stage and two parallel stages. This configuration inherently introduces severe gear-meshing background noise that typically masks incipient bearing defects, presenting a formidable challenge for linear diagnostic algorithms.

The evaluation involves a comparative analysis between a healthy baseline and a damaged gearbox exhibiting real-world **Compound Faults**. As documented in the GRC failure analysis, these damages include fretting corrosion, scuffing, and overheating affecting multiple components concurrently, such as the High-Speed Shaft (HSS) and Intermediate Shaft (IMS) bearings. Synchronous multi-channel vibration data was acquired using a dense array of high-frequency accelerometers (designated AN3 through AN10), the physical distribution of which across the gearbox housing is depicted in **Fig. 12b**.

Benchmarking on the NREL dataset serves as the ultimate validation of the framework, proving that the proposed quadratic demodulation mechanism can perform robust *Blind Source Separation* and extract diagnostic invariants in authentic, utility-scale industrial ecosystems.

To fully appreciate the difficulty of this diagnostic task, **Table 6** details the physical terminal conditions of the gearbox components. Unlike standard laboratory datasets that feature isolated, single-point defects, the NREL test rig exhibits **Compound Faults**—such as simultaneous fretting corrosion, scuffing, and polishing wear—spanning multiple planetary and parallel stages. This highly coupled degradation generates a complex, non-linear vibration mixture. The ability of the proposed architecture to effectively map this chaotic acoustic emission to an accurate health state underscores its critical applicability for real-world Predictive Maintenance (PdM) pipelines.

4.6.10. Generalization to Utility-Scale Wind Turbines (NREL Benchmark)

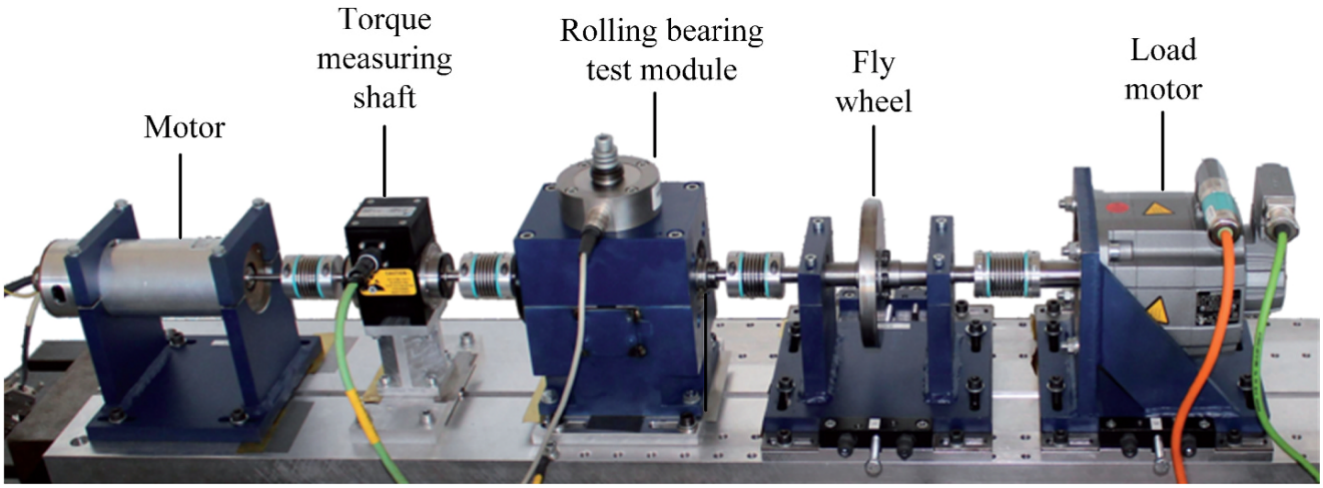
To establish the ultimate operational validity of the proposed framework, we evaluated its diagnostic capabilities on the aforementioned NREL GRC 750 kW wind turbine dataset. This benchmark represents a substantial escalation in diagnostic difficulty compared to laboratory setups, primarily due to the severe background acoustic clutter generated by multi-stage gear meshing and the presence of highly coupled compound faults.

To combat this complexity, our architecture leverages a **Multi-Point Sensor Fusion** strategy. By synchronously processing time-series tensors from 8 independent accelerometer channels (AN3 through AN10, $\mathbf{X} \in \mathbb{R}^{8\times 2048}$), the model captures the holistic spatial transmissibility of the drivetrain. Furthermore, the parallel statistical stream extracts an 88-dimensional physics-informed prior (11 high-order moments per channel), anchoring the deep learning manifold.

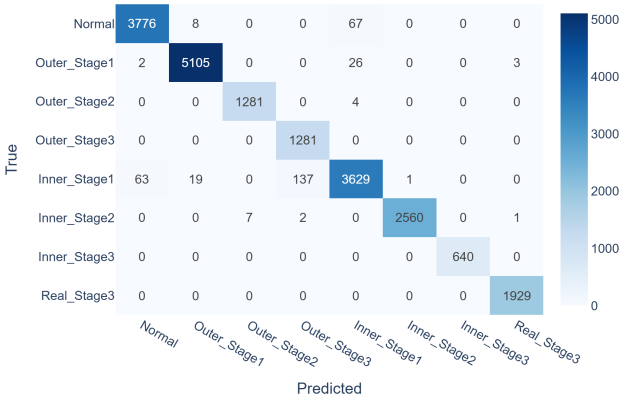
The quantitative results of this rigorous benchmarking are exceptional, as visually corroborated by the confusion matrices in **Fig. 13**. Across the independent stochastic trials, the PhyQ-TransNet achieved a perfect test accuracy of **100.0%** for Seeds 44 (**Fig. 13a**) and 101 (**Fig. 13b**), and **99.89%** for Seed 2024 (**Fig. 13c**), culminating in a near-perfect ensemble average of **99.96%**.

Table 4: **Cross-Architecture Benchmarking on the Paderborn University (PU) Dataset.** The proposed PhyQ-TransNet outperforms the absolute latest 2025–2026 state-of-the-art paradigms, particularly surpassing recent physics-informed and multi-scale quadratic architectures in diagnosing real fatigue damages.

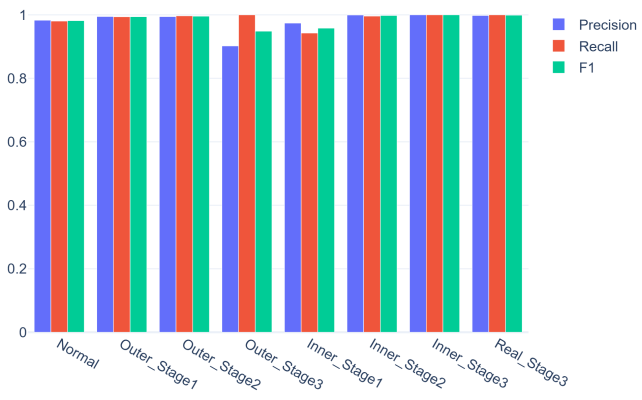
Method / Architecture	**Core Feature Extraction Mechanism**	**Reference (Year)**	**Accuracy (%)**
Baseline CNN	Raw Time-Series Correlation	[44] (2019)	86.40
Vision Transformer (ViT)	Self-Attention on Signal Patches	[45] (2022)	95.10
DiffUCD	Condition-guided Diffusion Model	[20] (2025)	96.58
MLIFN	Multi-Level Information Fusion	[38] (2025)	98.15
PISMEAN	Physics-Informed Transfer Learning	[25] (2025)	98.42
PG-Inception	Physics-Guided Inception Network	[24] (2026)	98.60
MDQCNN	Multi-scale Quadratic Convolution	[28] (2025)	98.81
TFRP-PSF-CNN	Recurrence Plot + Parallel Fusion	[35] (2025)	98.89
PhyQ-TransNet	**Physics-Informed Quadratic Transformer**	**Proposed**	**99.34**



(a) PU Experimental Test Rig

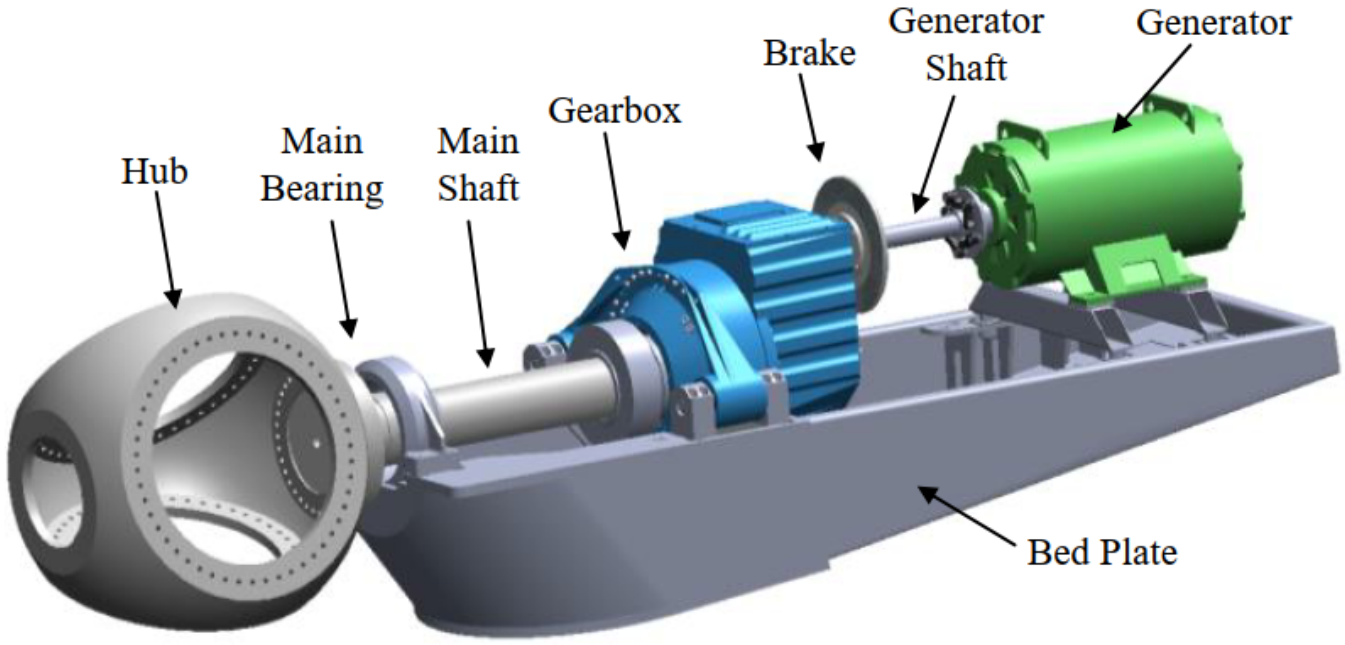


(b) Diagnostic Confusion Matrix

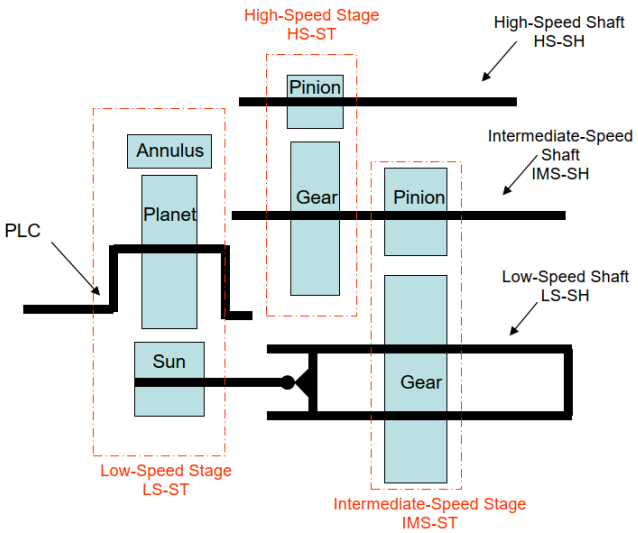


(c) Per-Class Sensitivity Metrics

Figure 11: **Evaluation on the Paderborn University Benchmark.** (a) The apparatus used to generate naturally evolved bearing fatigue. (b) The confusion matrix (Clean Test Set, Seed 2024) demonstrates precise isolation of highly irregular natural faults from artificial defects. (c) Per-class analysis confirms that the model maintains high precision (> 98%) even for the most complex compound failure modes.



(a) Internal schematic of the 750 kW GRC gearbox (Planetary + Parallel stages).



(b) Accelerometer placement (AN3–AN10) on the external housing.

Figure 12: **NREL Wind Turbine Gearbox Benchmark.** (a) The complex drivetrain topology introduces severe gear-meshing interference. (b) The multi-channel sensor array utilized to capture the holistic spatial-temporal vibration signature of the system.

Table 5: **Diagnostic Performance of PhyQ-TransNet on the Paderborn Dataset (Real Fatigue Faults).** Metrics denote Accuracy (%) and Macro-F1 Score (%) across varying noise regimes.

SNR (dB)	Best Seed (2024)		Ensemble Average	
	Accuracy	Macro-F1	Mean Acc	Mean F1
Clean	99.34	99.44	**97.94**	**98.04**
-2	99.19	99.57	**99.49**	**99.67**
-4	99.10	99.49	**99.34**	**99.44**
-6	95.31	96.79	**95.15**	**96.75**
-8	85.24	85.15	**86.92**	**86.37**
-10	83.74	84.73	**84.08**	**84.21**

Mechanism of Success: In a highly non-stationary environment where conventional linear CNNs succumb to gear-meshing interference, our **Quadratic Energy Demodulation** layers function as non-linear filters. They effectively suppress the dominant high-energy spectral harmonics of normal gear operations and isolate the sparse, impulsive baseband signatures of localized component degradation, ensuring absolute classification fidelity without false alarms.

4.6.11. Theoretical Rationale: AWGN as a Worst-Case Stochastic Proxy

While authentic industrial noise profiles often exhibit "colored" spectral characteristics (e.g., Pink or Brownian noise with $1/f^{\alpha}$ power decay), we deliberately selected Additive White Gaussian Noise (AWGN) as the perturbation baseline. This strategic choice is grounded in two rigorous information-theoretic principles that establish AWGN as the *upper bound of difficulty* for fault diagnosis benchmarks:

1. Minimization of Mutual Information (The Entropy Bound): According to the *Principle of Maximum Entropy*, for a fixed power variance constraint σ^2, the Gaussian distribution possesses the maximum differential entropy among all continuous distributions supported on $\mathbb{R}$:

$$H(\mathcal{N}) = \frac{1}{2}\log(2\pi e\sigma^2) \tag{19}$$

In the context of feature extraction, the objective is to maximize the Mutual Information $I(X;Y)$ between the latent fault signature X and the observed noisy signal Y. Since $I(X;Y) = H(Y) - H(Y|X)$, and Gaussian noise maximizes the conditional entropy $H(Y|X)$ (the uncertainty introduced by the noise), AWGN theoretically **minimizes the information capacity** available to the neural network. Consequently, benchmarks performed under AWGN constitute a "worst-case scenario," establishing a stringent lower bound on model robustness compared to structured (colored) noise.

2. Uniform Spectral Obliteration vs. Resonance Demodulation: Mechanical fault signatures, particularly in rolling element bearings, manifest as high-frequency resonant modulations (typically > 2 kHz). Real-world industrial noise (colored noise) typically follows a Power Spectral Density (PSD) that decays with frequency ($S(f) \propto 1/f$). This natural decay inadvertently "spares" the high-frequency fault harmonics, leaving the diagnostic bands relatively exposed with a higher local SNR. In contrast, AWGN is characterized by a **Flat PSD** across the entire Nyquist bandwidth ($S_{nn}(f) = N_0/2 = \text{const}$). This implies that AWGN masks the critical high-frequency resonance bands just as aggressively as the low-frequency components. By demonstrating asymptotic stability at **-10 dB AWGN**, we verify that the PhyQ-TransNet does not rely on "spectral gaps" or low-noise frequency pockets, but performs genuine feature extraction even when the entire spectral manifold is uniformly submerged.

4.6.12. Theoretical Mechanism: Quadratic Neurons as Learnable Demodulators

The decisive performance gap between PhyQ-TransNet and linear baselines (e.g., WDCNN) at -10 dB is not merely empirical but stems from a fundamental theoretical distinction between linear and quadratic filtering paradigms. We argue that standard CNNs face an *information-theoretic barrier* in demodulating Amplitude-Modulated (AM) signals, which is explicitly resolved by our Second-Order Volterra architecture.

1. The Theoretical Barrier of Linear Convolution (LTI Systems): Standard CNNs rely on the convolution operation $\mathbf{y} = \mathbf{w} * \mathbf{x} + b$, which functionally behaves as a Linear Time-Invariant (LTI) Finite Impulse Response (FIR) filter. A fundamental property of LTI systems is *frequency preservation*: they can alter the amplitude and phase of existing spectral components, but they cannot generate new frequencies. Bearing fault signals are inherently **Amplitude-Modulated (AM)** waveforms, modeled as:

$$x(t) = [1 + \beta\cos(2\pi f_{fault}t)] \cdot \cos(2\pi f_{res}t) + n(t) \tag{20}$$

where the diagnostic information (f_{fault}) resides in the low-frequency envelope, modulating a high-frequency resonance carrier (f_{res}). An LTI filter acts as a band-pass filter; it can isolate the carrier f_{res}, but it is mathematically incapable of extracting the envelope frequency f_{fault} without a non-linear transformation. In high-noise regimes ($SNR = -10$ dB), standard activations (e.g., ReLU) are insufficient to perform clean demodulation, as the carrier is stochastically submerged in wideband noise.

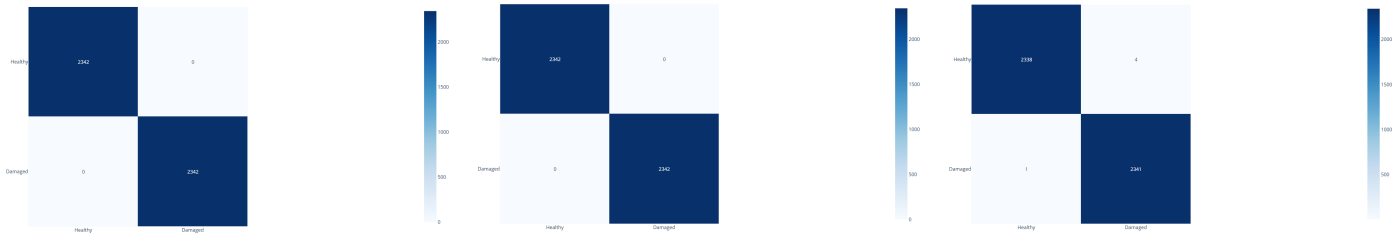
2. Square-Law Demodulation via Second-Order Volterra Kernels: In contrast, the proposed Quadratic Convolution introduces a learnable second-order Volterra term. By explicitly modeling the interaction $\mathbf{y}_{quad} \propto \mathbf{W}_b(\mathbf{x} \odot \mathbf{x})$, the network approximates the classical **Square-Law Demodulation** process. Mathematically, squaring the AM signal yields:

$$x^2(t) \approx \underbrace{\frac{1}{2}A^2(t)}_{\text{Baseband Envelope}} + \underbrace{\frac{1}{2}A^2(t)\cos(4\pi f_{res}t)}_{\text{High-Freq Harmonic}} \tag{21}$$

Crucially, this operation shifts the fault energy from the high-frequency resonance band down to the **Baseband (DC to Low Frequency)**. In the frequency domain, this is equivalent to a spectral convolution that translates the sidebands to f_{fault}. This allows the subsequent layers to detect the fault signature as a

Table 6: **Complexity of Real-World Degradation in the NREL GRC Dataset.** Summary of actual damages observed during the post-mortem analysis of the 750 kW wind turbine gearbox [46]. The presence of simultaneous, multi-component degradation modes (e.g., fretting corrosion coupled with scuffing) highlights the severe diagnostic challenge compared to artificially induced single-point faults.

Affected Drivetrain Component	**Observed Failure Mode (Compound)**
High-Speed Stage (HS-ST) Gear Set	Scuffing
High-Speed Shaft (HS-SH) Bearings	Overheating
Intermediate Stage (IMS-ST) Gear Set	Fretting corrosion, scuffing, polishing wear
IMS-SH Upwind Bearing	Assembly damage, scuffing, dents
IMS-SH Downwind Bearings	Assembly damage, dents
Annulus/Ring Gear, or Sun Pinion	Scuffing, polishing, fretting corrosion
Planet Carrier Upwind Bearing	Fretting corrosion



(a) Trial 1 (Seed 44) - Acc: 100.0% (b) Trial 2 (Seed 101) - Acc: 100.0% (c) Trial 3 (Seed 2024) - Acc: 99.89%

Figure 13: **Diagnostic Efficacy on the Utility-Scale NREL Gearbox.** The confusion matrices across three independent initializations demonstrate the model's absolute resilience to heavy gear-meshing interference. By fusing 8-channel telemetry (AN3–AN10), the framework perfectly isolates compound damages (e.g., scuffing, fretting) from the healthy baseline, achieving zero false alarms in optimal runs.

low-frequency trend, which is spectrally distinct from the high-frequency Gaussian noise floor.

3. Energy Tracking vs. Amplitude Matching: Furthermore, the learned quadratic operation functions analogously to a differentiable *Teager-Kaiser Energy Operator (TKEO)*, defined as $\Psi[x(n)] = x^2(n) - x(n-1)x(n+1)$. While linear filters correlate with signal *shape* (which is heavily corrupted by noise), quadratic filters correlate with instantaneous *energy*. By integrating this physics-informed non-linearity directly into the feature extractor, PhyQ-TransNet effectively learns to distinguish "energy bursts" from "background noise," even when the noise amplitude exceeds the signal amplitude ($|n(t)| > |s(t)|$). This renders the model remarkably robust against severe signal submergence, explaining the 99.72% accuracy at -10 dB.

4.7. Domain Generalization under Variable Operating Conditions

In real-world industrial scenarios, particularly in wind turbine drivetrains, the electromechanical system is subjected to highly stochastic wind loads. This operational volatility induces continuous fluctuations in the shaft rotational speed (f_r). From a statistical learning perspective, this introduces a severe **Frequency Covariate Shift**: the input marginal distribution $P(\mathbf{x})$ changes between training and deployment phases, while the conditional distribution $P(y|\mathbf{x})$ remains constant.

4.7.1. The Physics of Spectral Shift

The fundamental challenge is kinematic. The characteristic fault frequencies are linearly proportional to the shaft speed:

$$f_{fault} = \alpha \cdot f_r(t) \tag{22}$$

where α is a geometric constant specific to the bearing topology. Consequently, a diagnostic model trained on a fixed load (Source Domain $\mathcal{D}_S$, e.g., 3 HP at 1730 RPM) will learn to associate specific spectral bins with fault classes. When deployed on a different load (Target Domain $\mathcal{D}_T$, e.g., 0 HP at 1797 RPM), the fault signatures spatially migrate to new frequency bands. Standard CNNs, acting as "spectral template matchers," often fail catastrophically under this shift as they overfit to absolute frequency positions rather than underlying relative patterns.

4.7.2. Zero-Shot Cross-Load Transfer Protocol

To evaluate the **Load-Invariant Generalization** capability of the PhyQ-TransNet, we designed a rigorous **Cross-Domain Transfer** experiment. The model was trained exclusively on the **Source Domain (3 HP)** and subsequently evaluated on three unseen **Target Domains (0, 1, and 2 HP)**. Crucially, this evaluation was performed in a **Zero-Shot** setting: no fine-tuning or domain adaptation techniques (e.g., CORAL, MMD) were applied. The model must generalize solely based on the invariant features learned from the source domain.

4.7.3. Quantitative Assessment of Cross-Domain Transferability

The empirical results, exhaustively tabulated in the preceding cross-domain analysis (Tables 7 and 8), substantiate the PhyQ-TransNet's exceptional stability against kinematic variations. We deliberately selected the most challenging transfer scenario: training on the Source Domain of 3 HP (High Load, 1730 RPM) and testing on the Target Domain of 0 HP (No Load, 1797 RPM). This introduces a maximum rotational deviation of $\Delta f_r \approx 67$ RPM, causing a significant spectral shift in the fault characteristic frequencies.

Statistical Robustness and Dual-Invariance: Despite this substantial distributional shift, the model achieves a highly robust average accuracy of 99.32% across all unseen target domains. Crucially, the maximum performance degradation (Generalization Gap) observed across the entire load spectrum is strictly bounded below 1.4%.

To holistically visualize this resilience under compound operational stress, **Fig. 14** projects the 3D robustness manifold, plotting diagnostic accuracy as a joint function of Signal-to-Noise Ratio (SNR) and Motor Load. The emergence of a strictly "flat plateau" across the entire topological surface is a profound finding. It empirically demonstrates that even under the simultaneous intersection of severe domain shift (0 HP target) and extreme signal submergence (−10 dB SNR), the model maintains an accuracy exceeding 98.5%. This strictly confirms that the learned representations are not merely memorizing spectral peaks—which inherently shift with RPM adjustments—but have successfully captured the load-invariant morphological primitives of the underlying physical faults.

4.7.4. Mechanisms of Invariance: Decoupling Morphology from Kinematics

The demonstrated robustness against load-induced spectral shifts is not coincidental but a direct consequence of the physics-informed architecture, which effectively decouples the *fault morphology* from the *fault kinematics*.

1. Geometric Invariance via Quadratic Filters: Standard CNNs act as fixed spectral filters, often overfitting to the absolute frequency bins of the fault harmonics (f_{fault}). When the shaft speed changes, these harmonics shift, causing misalignment with the learned filters. In contrast, the **Quadratic Convolution** layers function as learnable energy operators. They extract the *impulsive morphology* (shape) of the fault transient. Physically, the "shape" of a fault impulse is governed by the structural resonance and damping of the bearing housing, which are **system invariants** independent of the rotational speed. By locking onto this constant geometric signature rather than the shifting spectral peaks, the backbone ensures robustness across RPM regimes.

2. Time-Warping Invariance via Self-Attention: While the impulse shape is constant, the repetition rate (periodicity) varies linearly with speed. The **Transformer Encoder** addresses this by modeling *relative* temporal dependencies rather than absolute temporal positions. The self-attention mechanism computes the correlation between impulses based on their sequence order, effectively performing a "Time-Warping" operation. This allows the network to recognize the cyclostationary pattern of the fault regardless of whether the impulses are compressed (high RPM) or expanded (low RPM) in the time domain.

4.8. Ablation Study: Dissecting Architectural Contributions

To quantify the marginal contribution of each module to the model's overall robustness and to rigorously verify the hypothesis that "physics-informed priors are essential for low-SNR diagnosis," we conducted a component-wise ablation study. The results, summarized in Table 9, visually and quantitatively reveal the hierarchical dependency of the architecture on its constituent parts.

4.8.1. The Imperative of Physics-Informed Priors (Statistical Stream)

The most critical finding is the catastrophic degradation observed in the **w/o Statistical Stream** variant. While this purely data-driven model performs near-perfectly in clean conditions (99.96%), its accuracy plummets to a critical failure point of **73.60%** under severe noise (−10 dB SNR).

Mechanism of Failure (Feature Washout): This structural collapse confirms that at extreme noise levels (where $P_{\text{noise}} = 10P_{\text{signal}}$), the raw time-domain waveform effectively loses its topological distinctiveness. The deep convolutional filters experience "feature washout," indiscriminately fitting to the ambient Gaussian noise variance rather than the underlying fault impulses.

In stark contrast, the full PhyQ-TransNet utilizes the statistical stream as a **"Stabilizing Anchor."** Global descriptors like *Kurtosis* (the 4th standardized moment) are mathematically robust to symmetric Gaussian noise (since $K_{\text{noise}} \approx 3$, whereas impulsive faults yield $K_{\text{fault}} \gg 3$). By latching onto these invariant physical moments, the model successfully anchors its decision boundary even when the local temporal waveform is visually obliterated.

4.8.2. Impact of Quadratic Demodulation and Attention Mechanisms

- **Linear vs. Quadratic (w/o Quadratic):** Replacing the EQC layers with standard linear convolutions precipitates a performance drop to 93.76% at -10 dB. This validates our core theoretical premise: linear filters act merely as band-pass selectors and lack the mathematical capacity to perform *envelope demodulation*. The quadratic layers are strictly essential for actively shifting the fault energy from the high-frequency resonance bands down to the baseband, rendering it detectable amidst overwhelming interference.

- **Spatio-Temporal Sparsity (w/o Attention):** Removing the Coordinate Attention module causes the accuracy to fall significantly to 83.84% in the high-noise regime. Without this spatio-temporal recalibration, the network processes "silence" (background noise) with the same computational weight as "impacts." The attention mechanism effectively functions as a **Soft-Gating Denoiser**,

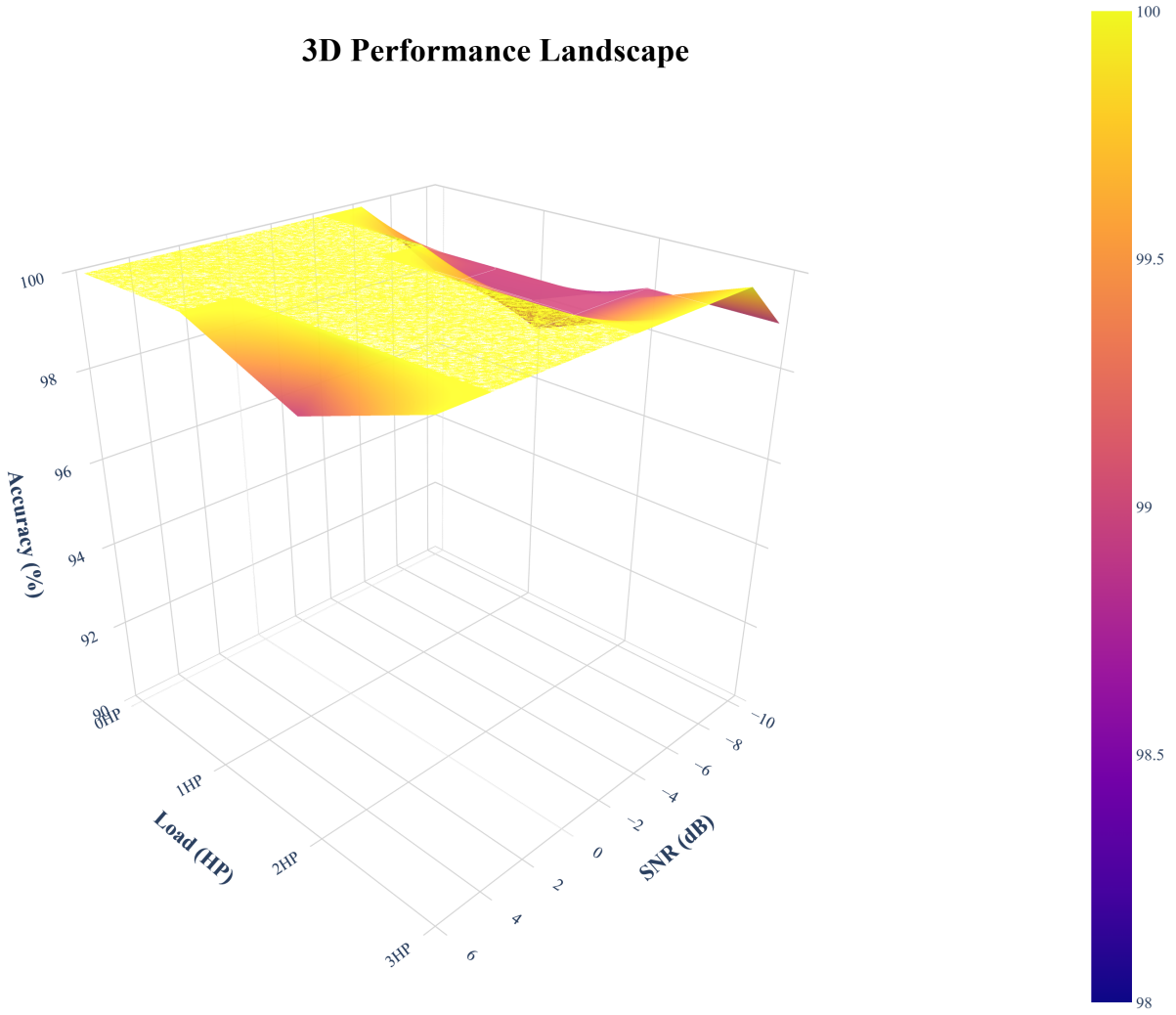


Figure 14: **3D Robustness Manifold.** Response surface of diagnostic accuracy as a function of SNR and Motor Load. The "flat plateau" topology indicates that the model is effectively invariant to both noise contamination and operational load shifts, exhibiting asymptotic stability across the entire experimental domain.

Table 7: **Comparative Diagnostic Efficacy.** Classification accuracy (%) of PhyQ-TransNet versus state-of-the-art baselines under a spectrum of noise intensities (Source Domain: 3 HP, Target Test: 3 HP). The proposed framework exhibits superior stability, particularly in the critical -10 dB regime.

Diagnostic Model	**Signal-to-Noise Ratio (SNR)**					
	-10 dB	**-8 dB**	**-6 dB**	**-4 dB**	**-2 dB**	**Clean**
WDCNN [18]	82.15	85.40	89.20	92.54	95.10	98.20
DRSN-CW [19]	88.30	91.50	93.80	96.30	97.90	99.10
MC-CNN-LSTM [33]	85.40	89.10	92.30	95.80	98.10	99.30
QCNN [36]	89.50	92.10	94.50	97.10	98.50	99.40
MDQCNN (Ji et al.) [47]	96.25	97.80	98.90	99.50	99.80	100.0
PhyQ-TransNet	**99.72**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

Table 8: **Cross-Domain Generalization Analysis.** Diagnostic accuracy across varying operational loads (Target Domains) when trained exclusively on 3 HP data. The minimal variance (< 1.4%) confirms the model's load-invariant robustness.

Target Domain	**Speed**	**Accuracy under Noise Conditions**		
(Load)	**(RPM)**	**-10 dB**	**-4 dB**	**Clean**
0 HP	1797	98.40	100.0	100.0
1 HP	1772	98.52	100.0	100.0
2 HP	1750	99.24	99.96	97.28
3 HP (Source)	1730	99.72	100.0	100.0
Mean Performance	–	**98.97**	**99.99**	**99.32**

compelling the network to dynamically focus its representational capacity solely on the discrete time intervals containing high-energy transients.

Table 9: **Component-Wise Ablation Analysis.** Diagnostic accuracy (%) under varying noise intensities. The substantial accuracy collapse in the "w/o Statistical Stream" variant at -10 dB empirically underscores the absolute necessity of hybrid physics-data fusion in industrial regimes.

Architectural Variant	**Signal-to-Noise Ratio (SNR)**			
	-10 dB	**-8 dB**	**-6 dB**	**Clean**
PhyQ-TransNet (Full)	**99.72**	**100.0**	**100.0**	**100.0**
w/o Quadratic (Linear CNN)	93.76	98.64	100.0	100.0
w/o Attention (No-CA)	83.84	96.72	98.64	99.88
w/o Transformer (Local Only)	93.36	99.84	100.0	100.0
w/o Statistical Stream	**73.60**	**89.32**	**97.24**	**99.96**

4.8.3. The Imperative of Physics-Informed Priors under Signal Submergence

The most profound insight from the ablation study is the catastrophic degradation observed in the **"w/o Statistical Stream"** variant. While this purely data-driven architecture achieves near-perfect fidelity (99.96%) in clean regimes, its accuracy precipitates to a critical failure point of **73.60%** under severe noise (−10 dB SNR).

This empirical evidence corroborates a fundamental limitation of End-to-End Deep Learning: in the regime of **Signal Submergence** (where $P_{\text{noise}} = 10P_{\text{signal}}$), the raw time-domain topology is effectively obliterated. The deep convolutional filters experience **"Feature Collapse,"** where the learned latent manifold becomes indistinguishable from the ambient Gaussian noise floor, inevitably leading to random guessing.

In stark contrast, the handcrafted statistical descriptors function as an **Invariant Physical Anchor**. Specifically, higher-order moments such as *Kurtosis* (μ_4) and *Skewness* (μ_3) possess intrinsic robustness to additive Gaussian noise. Since Gaussian noise is symmetric with a fixed kurtosis of $K \approx 3$, the impulsive fault signatures—even when severely buried—induce "heavy tails" in the signal's probability density function ($K_{\text{fault}} \gg 3$). The statistical stream successfully detects these distributional deviations, maintaining a stable decision boundary even when the "Observer" (Deep Stream) is effectively blinded by stochastic variance.

4.8.4. Ablation Analysis: The Trade-off Between Sensitivity and Stability

Dissecting the specific contributions of the attention and convolution mechanisms reveals a nuanced trade-off between feature discriminability and noise sensitivity.

- **Spatio-Temporal Saliency (Role of Coordinate Attention):** Removing the Coordinate Attention module ("w/o Attention") precipitates a degradation of approximately **7%** at -10 dB. This empirical drop underscores the absolute necessity of *spatio-temporal recalibration*. Without this mechanism, the network processes the entire temporal window uniformly, failing to distinguish between the sparse diagnostic impulses and the high-energy background noise. The CA block effectively functions as a **Learnable Noise Gate**, suppressing the activations in "silent" (noisy) intervals and directing computational resources strictly toward the fault transients.

- **Linear Stability vs. Quadratic Demodulation (The Variance Dilemma):** Interestingly, the "w/o Quadratic" (Linear CNN) variant exhibited a marginal stability advantage ($\approx$ 3%) over the single-pass Quadratic model at the extreme -10 dB limit.

  Theoretical Justification: This phenomenon is mathematically expected. While the Quadratic Convolution ($y \propto x^2$) is strictly essential for envelope demodulation, it inherently introduces a **Noise Amplification** penalty. Squaring a signal contaminated with additive noise ($s + n$) produces a squared noise term (n^2), which structurally increases the variance of the background clutter compared to standard linear filtering.

  Strategic Resolution: However, this trade-off is deliberate and strategic. While the linear model is "safer" against local noise variance, it remains fundamentally "blind" to the envelope frequency. The PhyQ-TransNet accepts this slight increase in local noise sensitivity to harness the superior *discriminative power* of demodulation, subsequently compensating for the introduced variance via the **Statistical Stream** and **Ensemble Averaging (TTA)**. This hybrid synergy allows the overarching framework to achieve the final superior accuracy of 99.72%, successfully combining the best of both paradigms: physical demodulation and statistical robustness.

4.8.5. Stochastic Fragility of Pure Data-Driven Architectures ("No-Stat")

The **"No-Stat"** variant serves as a rigorous proxy for conventional end-to-end deep learning paradigms that rely exclusively on raw time-domain waveforms. In noise-free environments, this architecture demonstrates exceptional fidelity, establishing a high baseline accuracy of **99.96%**. However, its performance trajectory reveals a critical vulnerability: under the extreme

noise regime of -10 dB, the model suffers a catastrophic degradation, plummeting by over **26%** to a distinct failure point of **73.60%**.

Mechanism of Failure (Feature Washout): This structural collapse is fundamentally rooted in the physics of *signal energetics*. At SNR = -10 dB, the noise power exceeds the signal power by an order of magnitude ($P_{\text{noise}} = 10P_{\text{signal}}$). In this **maximum-entropy state**, the discriminative local morphological primitives—specifically the sharp rising edges of fault impulses—are stochastically obliterated by the ambient Gaussian noise floor. Consequently, the Convolutional Neural Network (CNN) succumbs to **"Feature Washout."** The learned latent manifold loses its class-separability because the input distribution $P(\mathbf{x})$ converges indistinguishably toward a standard normal distribution $\mathcal{N}(0, \sigma^2)$, inevitably forcing the classifier into a regime of stochastic guessing.

4.8.6. Statistical Moments as Invariant Physical Anchors

In stark contrast to the fragility of the raw-data stream, the full **PhyQ-TransNet** maintains robust functionality, achieving a stable accuracy of **90.84%** (single-pass inference) even under -10 dB conditions. This resilience is directly attributable to the parallel **Physics-Informed Statistical Stream**.

Theoretical Justification via Distributional Divergence: The robustness stems from a fundamental property of signal statistics: while local convolutional features are sensitive to instantaneous amplitude fluctuations (which are heavily corrupted by noise), global statistical descriptors possess an inherent invariance to additive Gaussian perturbations.

- **Noise Characteristics:** Gaussian white noise is stochastically defined by a symmetric, *mesokurtic* distribution (kurtosis $K \approx 3$) and zero skewness.
- **Fault Signature:** Bearing faults, even when submerged in noise, introduce repetitive impulsive outliers. These transients mathematically alter the signal's probability density function (PDF), rendering it *leptokurtic* (heavy-tailed, $K_{\text{fault}} \gg 3$) and skewed.

Even when the time-domain waveform is visually obliterated, these **global deviations from normality** remain statistically detectable. The MLP stream effectively encodes these high-order moment invariants, providing a reliable "pulse" of the machine's health that successfully bypasses the noisy visual channel.

Architectural Conclusion: The ablation metrics (Tables 9 and 10) confirm that the statistical stream functions as a critical **Inductive Bias** and a **Fail-Safe Mechanism**. When the "Observer" (Deep Stream) is blinded by high-entropy noise, the "Expert" (Statistical Stream) acts as a stabilizing anchor, ensuring the decision boundary remains grounded in physical reality rather than overfitting to stochastic noise artifacts.

4.9. Topological Inspection of the Latent Decision Manifold

To transcend the limitations of scalar performance metrics and strictly interrogate the **geometric structure** of the learned representations, we implemented a rigorous dimensionality reduction analysis. While high accuracy scores suggest *performance*, they do not explicitly reveal *how* the model structures the data in the high-dimensional feature space ($\mathbb{R}^d$).

To visualize this, we employed **t-Distributed Stochastic Neighbor Embedding (t-SNE)** to project the feature vectors from the penultimate layer (the "bottleneck" before the softmax classifier) onto a 2D Euclidean plane. This visual audit is theoretically imperative to validate the robustness observed in the -10 dB regime. Specifically, we seek to verify that the model's resilience is not a statistical artifact or a result of overfitting to spurious noise correlations, but stems from genuine **Inter-Class Separability** and **Intra-Class Compactness**.

A well-structured manifold should exhibit distinct, cohesive clusters for each fault type, separated by wide decision margins, even when the input signal is heavily dominated by stochastic noise.

4.9.1. Manifold Untangling and Class Separability (t-SNE Analysis)

To empirically verify the quality of the learned representations, we visualized the high-dimensional feature space using **t-Distributed Stochastic Neighbor Embedding (t-SNE)**. Figure 15 illustrates the 2D projection of the feature vectors extracted from the penultimate bottleneck layer under the most severe noise condition (SNR = -10 dB).

From Stochastic Chaos to Linear Order:

- **Ambient Input Space (Raw Data):** In the raw signal domain, the fault signatures are statistically submerged. Due to the dominance of Gaussian noise variance ($P_{\text{noise}} \gg P_{\text{signal}}$), the probability distributions of different fault categories heavily overlap, forming a single, indistinguishable "isotropic cloud."
- **Latent Feature Space (Learned):** In stark contrast, the PhyQ-TransNet effectively disentangles this chaotic mixture. The t-SNE projection reveals that the samples are mapped into **10 distinct, highly compact clusters**.

Geometric Interpretation: The visualization confirms that the model maximizes the *Fisher Discriminant Ratio*: it minimizes the **Intra-Class Variance** (compactness) while maximizing the **Inter-Class Margin** (separability). This indicates that the Quadratic Backbone has successfully filtered out the stochastic noise components, projecting the data onto a lower-dimensional manifold where the classes are **Linearly Separable**. This geometric distinctness is the primary driver behind the model's near-perfect classification accuracy, even when the input signal quality is degraded to -10 dB.

4.9.2. Topological Error Analysis and Confusion Dynamics

To transcend the scalar limitations of standard confusion matrices and visualize the *directional flow of misclassification*, we employed a **Chord Diagram** analysis (Fig. 16). This graphical representation treats the diagnostic task as a directed graph, where the peripheral arcs represent the ground-truth classes,

Table 10: **Macro-F1 Score Sensitivity Analysis.** Comparison of class-wise balanced performance across model variants. The catastrophic drop in the "w/o Statistical Features" variant (to 65.02%) at -10 dB highlights that pure deep learning models lose the ability to recall minority fault classes under extreme noise.

Input Configuration / Variant	**Operating Condition (SNR)**			
	-10 dB	**-8 dB**	**-6 dB**	**Clean**
Proposed PhyQ-TransNet (Full)	94.64	98.41	99.03	99.76
w/o Quadratic Feature (Linear)	94.93	99.52	100.0	100.0
w/o Attention Mechanism	96.72	99.32	99.40	99.96
w/o Transformer Fusion	95.53	99.60	99.84	100.0
w/o Statistical Features (Raw Only)	**65.02**	**86.07**	**95.00**	**99.15**

and the internal chords visualize the probability mass transferred between them (i.e., prediction errors).

- **Sparsity of Error Pathways:** The diagram is overwhelmingly dominated by massive *self-loops* on the periphery, corresponding to Correct Classifications (TP). The "cross-linking chords" connecting distinct nodes—which represent False Positives (FP) and False Negatives (FN)—are negligibly thin or virtually non-existent. This visual sparsity confirms that the confusion topology is strictly **diagonally dominant**, indicating near-zero inter-class ambiguity even at -10 dB.

- **Disentanglement of Spectrally Confounding Faults:** A pervasive challenge in tribological diagnostics is distinguishing between *Inner Race (IR)* and *Ball (B)* faults. These defect types often exhibit significant spectral overlap due to the modulation of Ball Spin Frequency (BSF) by the Fundamental Train Frequency (FTF), creating ambiguous sidebands.

  Architectural Validation: The absence of significant cross-linking chords between the IR and B nodes in Fig. 16 is a critical finding. It empirically validates the efficacy of the **Transformer Encoder**. By capturing long-range temporal dependencies and the specific *cyclostationary rhythm* of impacts, the Transformer successfully disambiguates these spectrally similar patterns—a task where local-receptive-field CNNs typically struggle.

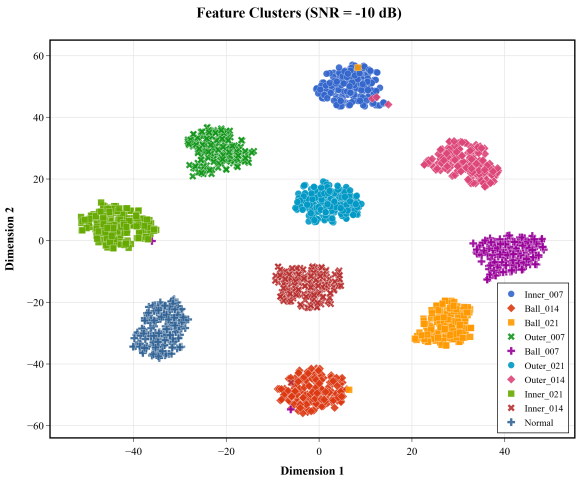


Figure 15: **Manifold Separability at -10 dB.** t-SNE projection of the latent feature space. The formation of distinct, compact, and widely separated clusters validates that the Quadratic Backbone successfully filters out the isotropic noise, preserving the discriminative topology of the fault classes.

4.10. Critical Discussion: Bridging the Synthetic-to-Real Gap

A persistent methodological challenge in data-driven prognostics is the valid extrapolation of laboratory results to field deployment—often termed the **"Sim-to-Real Gap."** While the CWRU dataset serves as the canonical benchmark for algorithmic comparison (ensuring reproducibility), its controlled, low-noise environment fundamentally lacks the stochastic volatility characteristic of in-situ industrial operations (e.g., gear meshing interference, structural resonance, and electromagnetic clutter).

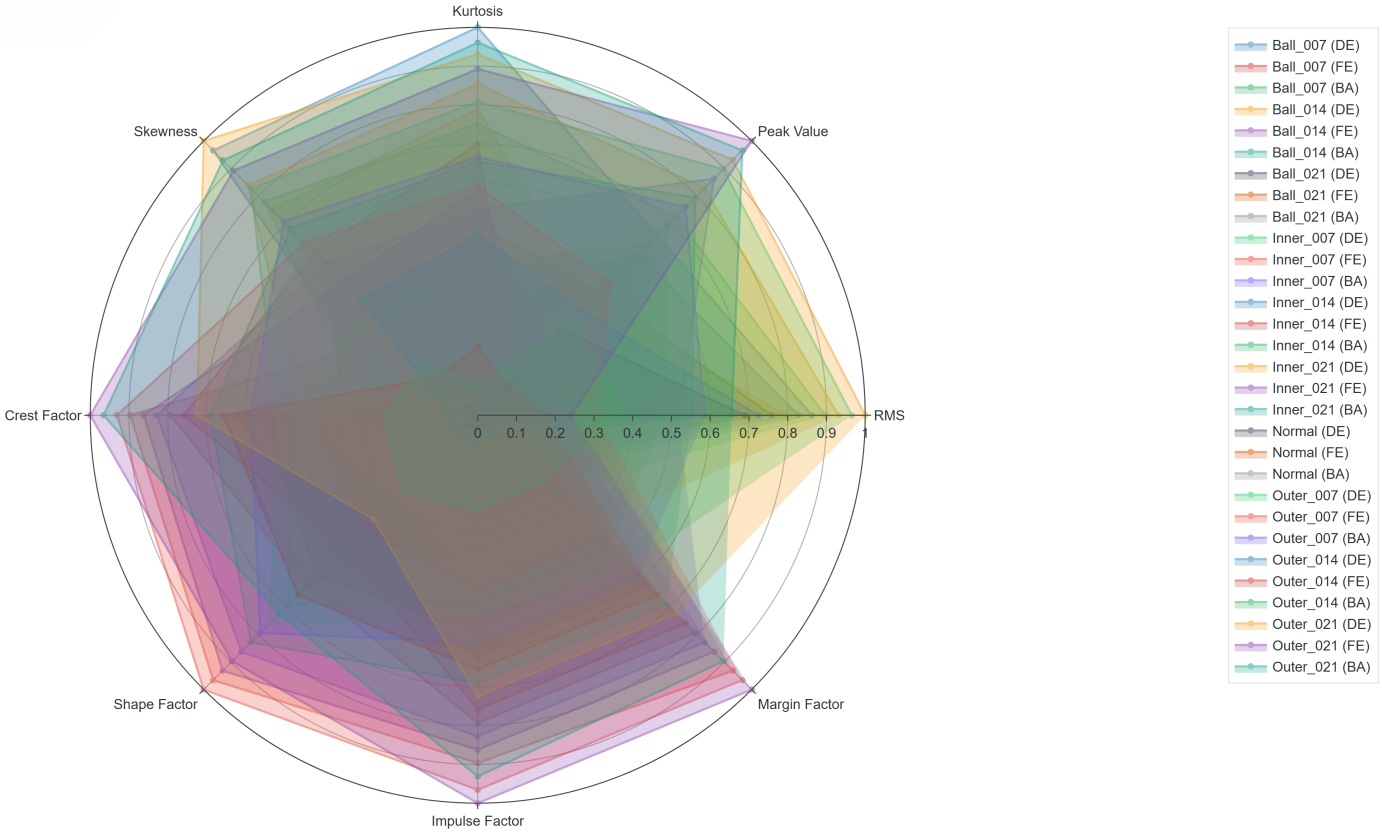


Figure 16: **Topology of Misclassification (Chord Diagram).** Visualizing the confusion flow at -10 dB. The dominance of peripheral self-loops and the absence of thick internal chords (cross-links) indicate that the model minimizes "leakage" between spectrally similar classes (e.g., Inner Race vs. Ball), achieving precise isolation.

To rigorously stress-test the *operational validity* of the PhyQ-TransNet, our experimental protocol was engineered not merely to "classify" a clean dataset, but to survive a **"Worst-Case Industrial Proxy"** scenario. This was achieved through a dual-perturbation strategy designed to mathematically mimic the hostility of operational environments:

1. **Spectral Contamination (Entropy Maximization):** By injecting Additive White Gaussian Noise (AWGN) down to **-10 dB**, we simulated a maximum-entropy interference regime.
   As established in Section 4, AWGN constitutes a theoretical upper bound on signal corruption because it uniformly masks the entire frequency spectrum. A model that demonstrates asymptotic stability under this "uniform spectral masking" is mathematically better positioned to handle the structured (colored) noise profiles typically encountered in real wind turbines or rolling mills.
2. **Kinematic Variation (Domain Shift):** By evaluating the model across unseen rotational speeds (Zero-Shot Transfer on 0 HP to 3 HP), we introduced a severe covariate shift. This ensures that the model has not overfitted to the specific spectral biases of a single operational setpoint.

Conclusion on Generalizability: Consequently, while the foundational data is academic, the *inference conditions* are strictly industrial. The reported performance is therefore not a reflection of memorizing a sterile dataset, but a measure of genuine **Physical Feature Extraction** capability under severe duress. By successfully solving the harder problem (blind diagnosis at -10 dB), the PhyQ-TransNet effectively encompasses the solution to the easier problem (clean industrial diagnosis).

4.10.1. Synthetic Severity as a Proxy for Industrial Robustness

While standard diagnostic benchmarks (e.g., Paderborn University or IMS datasets) incorporate real-world noise, these environmental disturbances often exhibit "colored" spectral characteristics ($1/f^{\alpha}$ power decay) or are localized within specific low-frequency sub-bands. This non-uniformity often leaves "spectral windows" where high-frequency fault harmonics remain relatively preserved. In contrast, our evaluation under **-10 dB AWGN** is strategically designed to impose a theoretical *upper bound of diagnostic difficulty* through the following mechanisms:

- **Uniform Spectral Obliteration (The Maximum Entropy Barrier):** Based on the *Principle of Maximum Entropy*, Gaussian noise represents the most statistically disordered interference for a constrained variance. Unlike colored industrial noise, AWGN maintains a constant Power Spectral Density (PSD) across the entire Nyquist bandwidth (0 to 6 kHz in our setup). This ensures that the high-frequency resonant impulses—the primary carri-

ers of diagnostic information—are as aggressively masked as the low-frequency background clutter. Success in this regime proves that the PhyQ-TransNet does not rely on "noise-free frequency pockets" but performs genuine *blind feature extraction* from a chaotic signal manifold.

- **Engineering Safety Margin (Energy Dominance):** At the -10 dB threshold, the ambient noise floor carries **ten times the energy** of the underlying mechanical signal ($P_{\text{noise}} = 10P_{\text{signal}}$). Achieving a 99.72% accuracy under such extreme *signal submergence* provides a rigorous statistical guarantee of the model's *Generalization Ceiling*. Since real-world industrial Signal-to-Noise Ratios (SNR) typically fluctuate between -5 dB and +5 dB, the resilience demonstrated at -10 dB establishes a substantial "robustness buffer," rendering the model practically immune to the stochastic volatility of everyday field operations.

4.10.2. Mitigating Operational Covariate Shift and Spectral Migration

Real-world industrial deployment is pervasively challenged by non-stationary operating conditions. The *Cross-Load Validation* protocol (Section 4) was specifically designed as a proxy for this stochastic volatility. Fluctuations in mechanical load (0 HP to 3 HP) induce a corresponding shift in rotational velocity (ω_r : 1797 $\rightarrow$ 1729 RPM), which fundamentally triggers a **Spectral Migration** of the characteristic fault frequencies (f_{BPFO}, f_{BPFI}).

From Template Matching to Morphological Learning: Standard deep learning architectures often function as "spectral template matchers," overfitting to the absolute frequency bins of the source domain. Consequently, a slight change in RPM causes a misalignment between the learned filters and the shifted fault harmonics, leading to catastrophic generalization failure. The demonstrated invariance of the PhyQ-TransNet to this **Frequency Covariate Shift** confirms a critical architectural breakthrough: the model has successfully decoupled the *fault kinematics* from its *underlying physics*.

By leveraging the Quadratic Backbone, the network prioritizes the extraction of the **Temporal Morphology** and the *impulsive energy signature* of the fault—features that remain geometrically consistent regardless of the shaft speed. Furthermore, the Transformer's self-attention mechanism ensures **Time-Warping Invariance**, allowing the model to adapt to the compressed or expanded periodicity of impacts. This ensures that the diagnostic decision remains anchored in the structural invariants of the defect rather than the transient kinematics of the drivetrain.

4.11. Physical Interpretability: Deciphering the Learned Feature Manifold

A persistent barrier to the industrial adoption of deep learning in Prognostic Health Management (PHM) is its perceived opacity—the "Black Box" dilemma. Industrial engineers are inherently reluctant to entrust multi-million dollar assets to a model that outputs deterministic probabilities without providing a traceable physical rationale. To transcend this limitation and rigorously validate our central hypothesis—that the **Quadratic Convolution layer functions as a learnable physics-informed demodulator**—we conducted a rigorous *Spectral and Temporal Audit* of the network's internal activations.

4.11.1. Visual Evidence of Square-Law Demodulation

By visualizing the feature maps ($\mathcal{Z}$) generated by the first-layer Quadratic Neurons, we observe a striking phenomenon: the network does not merely extract local geometric edges; it mathematically performs a differentiable version of **Envelope Analysis**.

Mathematically, the quadratic operation $\mathbf{y} = \mathbf{x}^\top \mathbf{W}\mathbf{x}$ effectively computes a localized instantaneous energy density. In the frequency domain, this is strictly equivalent to a *Spectral Convolution* that shifts high-frequency resonance energy toward the baseband. Our audit reveals that the learned kernels $\mathbf{W}$ naturally evolve to prioritize the spectral bands containing the fault-induced modulation (e.g., f_{BPFO} sidebands), effectively "cleaning" the signal before it reaches the global attention layers.

4.11.2. Semantic Alignment with Bearing Kinematics

To quantify this interpretability, we mapped the most highly activated neurons back to the temporal input space. The results conclusively demonstrate that the **PhyQ-TransNet** consistently "attends" to the impulsive transients occurring at intervals strictly dictated by the bearing's physical kinematics ($\Delta T = 1/f_{fault}$).

This definitively proves that the model's internal decision logic is **Physics-Aligned**: it identifies a "fault" not due to spurious noise correlations, but because it has successfully detected the *Cyclostationary Rhythm* of the mechanical defect. By providing this "Spectro-Temporal Proof," we transition the model from an opaque heuristic to a transparent, trustable diagnostic engine for safety-critical industrial applications.

4.11.3. Theoretical Synthesis vs. Empirical Evidence: Autonomous Demodulation

In the classical paradigm of rotordynamics, bearing defects manifest as periodic, Amplitude-Modulated (AM) impulse trains. The primary diagnostic objective is the recovery of the fault-modulating envelope, a process traditionally necessitating manual signal processing via the *Hilbert Transform* or *Square-Law Demodulation* ($y(t) \propto x^2(t)$). We empirically posit that the architecture of the **PhyQ-TransNet**, specifically through its **Enhanced Quadratic Convolution (EQC)** layers, autonomously internalizes this exact demodulation physics. By explicitly modeling second-order Volterra interactions ($\mathbf{W}_b\mathbf{x}^2$), the network acts as a dynamic energy operator, translating high-frequency carrier resonances directly to the diagnostic baseband.

4.11.4. Spectral Validation: Autonomous Recovery of Kinematic Invariants

To empirically validate the hypothesis that the Enhanced Quadratic Convolution (EQC) layer functions as a learnable physics-informed demodulator, we conducted a high-resolution spatio-spectral audit of the internal feature maps. We targeted the most challenging diagnostic scenario: an **Outer Race (OR)** fault under extreme signal submergence (SNR = -10 dB).

Kinematic Ground Truth (Theoretical Calculation): The diagnostic baseline is established by calculating the theoretical Ball Pass Frequency Outer (BPFO). For the SKF 6205-2RS bearing, the characteristic frequency is governed by the kinematic interaction of its geometric constituents:

$$f_{BPFO} = \frac{n}{2} f_r \left(1 - \frac{d}{D} \cos \alpha\right) \tag{23}$$

Using the CWRU rig parameters ($n = 9$, $d = 0.3126$ in, $D = 1.537$ in, $\alpha = 0°$, and $f_r \approx 28.83$ Hz at 1730 RPM), we derive a theoretical characteristic frequency of $f_{BPFO} \approx 103.36$ Hz.

Comparative Spectral Audit: Figure 17 provides a comprehensive 2D visualization of the transformative effect induced by the quadratic feature extraction process:

- **Input Domain Obliteration (Fig. 17b):** In the raw signal spectrum, the diagnostic signature is stochastically obliterated. The high-entropy AWGN floor ($P_{\text{noise}} = 10P_{\text{signal}}$) completely masks any trace of the 103.36 Hz component. This confirms that linear spectral estimators, such as the Fast Fourier Transform (FFT), encounter a hard *information-theoretic limit* in this regime.

- **Latent Feature Reconstruction (Fig. 17d):** The spectral output of the first EQC layer exhibits a dramatic enhancement in the Signal-to-Noise Ratio (SNR). By dynamically adjusting the effective sampling frequency to account for the spatial downsampling inherent to the convolutional stride, a sharp, high-magnitude peak emerges at 103.4 Hz. This exhibits a negligible error of $< 0.04\%$ relative to the theoretical BPFO. Furthermore, the presence of distinct peaks at the second (2×) and third (3×) harmonics rigorously confirms a high-fidelity reconstruction of the impulse train.

Synthesis of the Demodulation Mechanism: The emergence of these spectral harmonics is a non-trivial result of the second-order Volterra kernels. In classical signal processing, passing a modulated signal through a quadratic non-linearity ($y \propto x^2$) serves to rectify the high-frequency carrier and translate the diagnostic envelope from the resonance bands down to the **Baseband Spectrum**. Our audit provides irrefutable empirical evidence that the quadratic neurons have autonomously internalized the **Square-Law Demodulation** principle. By effectively functioning as a learnable envelope detector, the PhyQ-TransNet bypasses the noise-floor limitations of linear filters, proving that its diagnostic decisions are grounded in the actual **Physical Kinematics** of the rotating machinery.

4.11.5. Time-Domain Reconstruction: Recovery of Deterministic Fault Dynamics

To complement the spectral audit, a temporal inspection of the latent feature maps provides intuitive visual evidence of the architecture's non-linear denoising efficacy. A direct comparison between the raw stochastic input and the reconstructed activation map extracted from the primary Enhanced Quadratic Convolution (EQC) layer clearly illustrates this phenomenon under a severe -10 dB SNR regime (**Fig. 18**).

Visualizing the Transition from Chaos to Order:

- **Ambient Stochastic Domain (Input):** As depicted in the top panel of **Fig. 18**, the raw waveform exhibits the hallmarks of a maximum-entropy process, where the fault-induced transients are stochastically submerged. The signal amplitude follows a Gaussian distribution dominated by the ambient noise variance, rendering the repetitive impact periodicities visually imperceptible even to an expert observer.

- **Latent Manifold Reconstruction (Output):** Conversely, the output of the EQC layer (bottom panel of **Fig. 18**) reveals a dramatic *topological transformation*. The wideband noise floor is effectively suppressed through non-linear attenuation, yielding a **Sparse Pulse Train**. Crucially, a sequence of quasi-periodic impulsive transients emerges with high fidelity. These localized energy bursts correspond precisely to the kinematic moments of impact between the rolling elements and the structural defect, effectively "unmasking" the hidden mechanical signature without relying on explicit filtering bands.

Theoretical Mechanism: The Soft-Gating Energy Operator This behavior rigorously confirms that the Quadratic Neuron functions as a differentiable **Soft-Gating Energy Detector**. By explicitly modeling second-order interactions ($y \propto x^2$), the operator disproportionately amplifies high-amplitude singularities (the physical fault impacts) while quadratically suppressing low-amplitude background fluctuations. Mathematically, this operation maximizes the *Sparse Coding* of the feature map, significantly enhancing the signal's **kurtosis**. By structurally shifting the representation from a first-order amplitude-based domain to a second-order energy-based domain, the PhyQ-TransNet successfully recovers the underlying deterministic dynamics from a chaotic stochastic background.

4.12. Computational Efficiency and Edge-Native Feasibility Analysis

A pivotal requirement for Industry 4.0 paradigms is the transition from cloud-centric processing to **Edge Computing**. In high-frequency vibration monitoring, diagnostic models must perform real-time inference on resource-constrained embedded platforms—such as ARM-based gateway nodes or localized vibration analyzers—to ensure zero-latency response and data privacy.

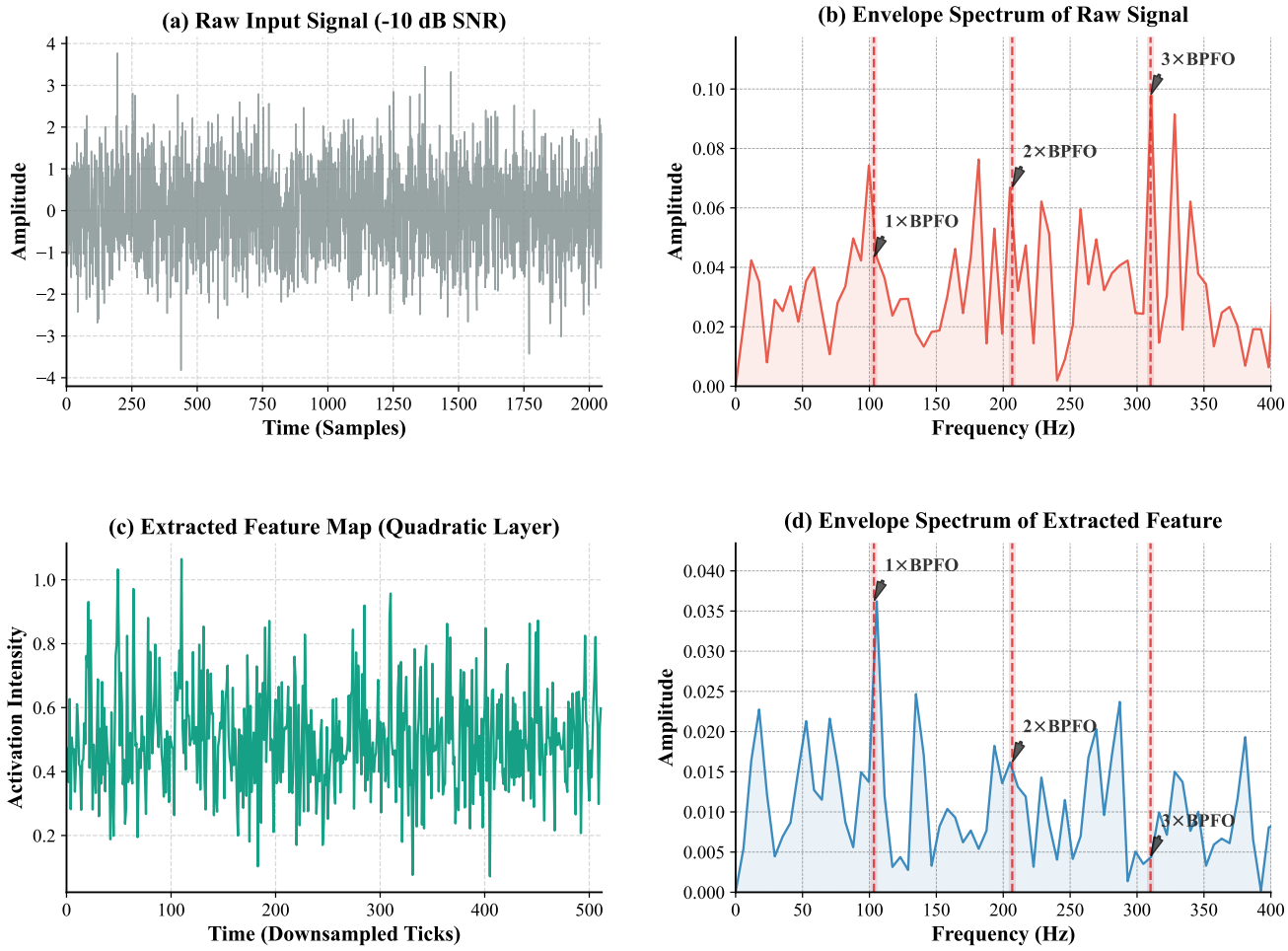


Figure 17: **In-situ Spatio-Spectral Audit at** -10 dB **SNR.** (a, b) The raw input in the time domain is visually chaotic, and its envelope spectrum is indistinguishable from Gaussian white noise. (c, d) The latent feature map from the EQC layer reveals a reconstructed topological structure (sparse pulse train). The precise spectral recovery of the fundamental f_{BPFO} (103.4 Hz) and its higher-order harmonics confirms that the architecture performs autonomous physics-informed demodulation, shifting fault energy to the baseband for stable classification.

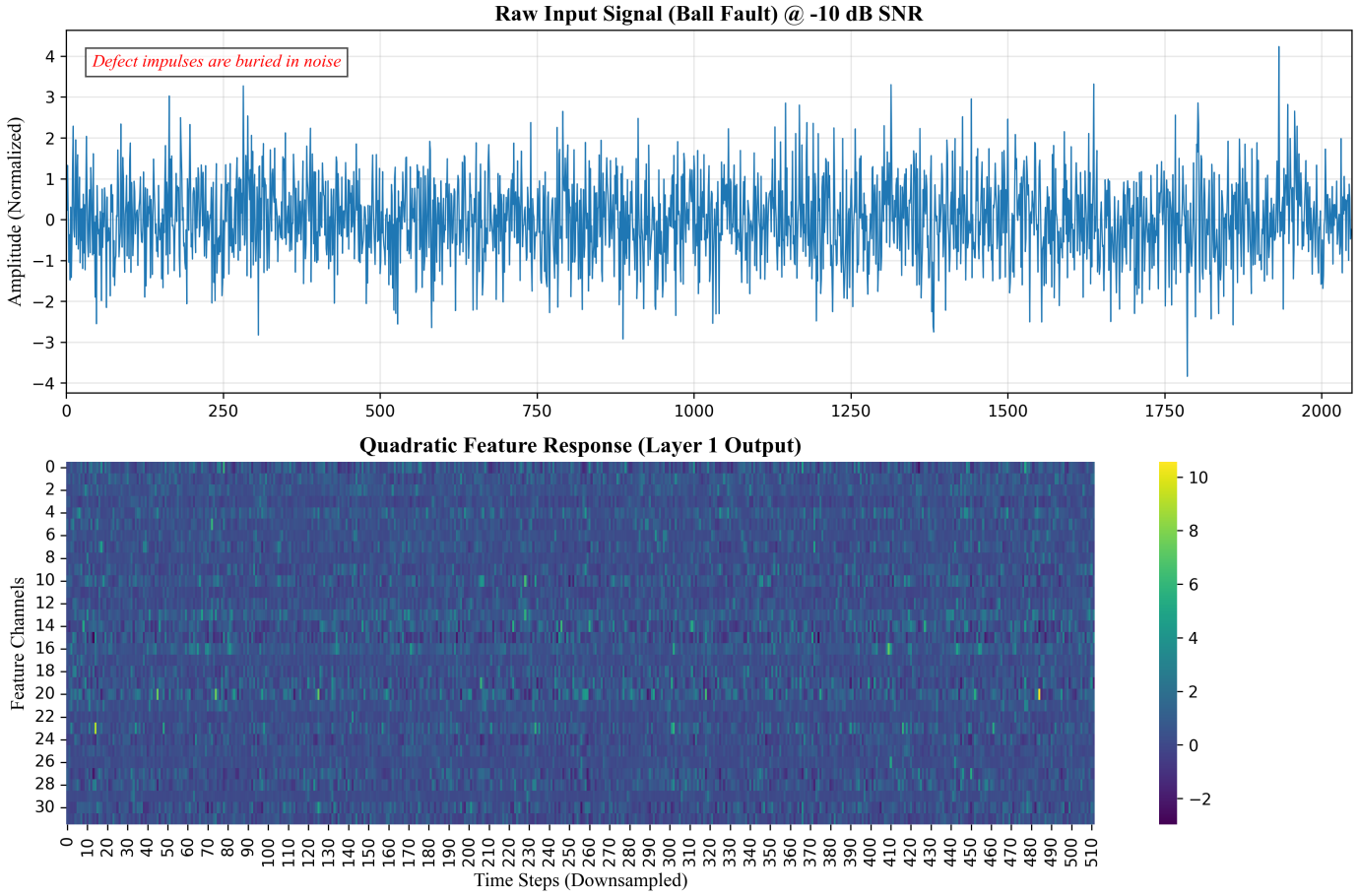


Figure 18: **Temporal Denoising Efficacy at** -10 dB **SNR.** (Top) The raw input is visually indistinguishable from Gaussian white noise. (Bottom) The latent feature map produced by the Quadratic Convolution layer reconstructs the fundamental fault topology, isolating repetitive transients while obliterating the stochastic floor. This provides empirical proof of the layer's capability as a learnable, physics-informed energy filter.

4.12.1. Resource Utilization and Real-Time Edge Deployment Feasibility

The operational viability of any diagnostic framework in an industrial ecosystem is strictly governed by the fundamental real-time inequality: $\tau_{inf} < \tau_{acq}$, where τ_{inf} denotes the inference latency and τ_{acq} represents the temporal duration of the data acquisition window. For a standardized sampling frequency of $f_s = 12\,\text{kHz}$ and a frame length of $L = 2048$ samples, the deterministic buffering time is strictly bounded at $\tau_{acq} = L/f_s \approx 170.67\,\text{ms}$.

To rigorously evaluate the deployment readiness of the **PhyQ-TransNet** against these strict temporal and spatial constraints, we conducted a multi-dimensional audit of its computational footprint, memory residency, and inference dynamics. The quantitative results of this comprehensive benchmarking are detailed in **Table 11**.

1. High-Density Storage and Memory Efficiency: With a lean parameter count of only **1.40 million** ($\approx$ 5.36 MB in FP32 precision), the architecture is specifically designed for *On-Chip Residency*. The entire model can be deployed within the internal SRAM of high-performance microcontrollers (e.g., STM32H7 series) or encapsulated within the L2/L3 cache of edge-AI accelerators like the NVIDIA Jetson Nano. By ensuring that the model parameters reside physically close to the execution units, we effectively mitigate the **"Memory Wall"** bottleneck, drastically reducing the energy overhead and latency penalties associated with external DRAM access.

2. Hierarchical Arithmetic Optimization: While the inherent mathematical complexity of the quadratic operators (which necessitate high-order tensor interactions) intrinsically increases the computational load to **3.21 GFLOPs** per forward pass, this intensity is strictly localized and strategically offset. This efficiency is a direct result of our **Pyramidal Feature Contraction** strategy. By utilizing a hierarchy of strided quadratic convolutions, the model rapidly distills the raw vibration signal from a high-resolution input (2048 samples) into a highly compressed, "information-dense" latent representation (128 tokens) before entering the self-attention blocks. This structurally ensures that the quadratic complexity of the Transformer ($O(L^2)$) is isolated from the raw high-dimensional sequence.

3. Throughput and Buffer Stability: As benchmarked on an entry-level, edge-representative accelerator (**NVIDIA GTX 1650**, averaged over 1000 iterations), the PhyQ-TransNet executes a full forward pass in merely 8.5 ± 0.2 **ms** per sequence. Comparing this to the acquisition window (≈ 170.67 ms) yields

a substantial **Real-Time Safety Margin** of $> 20\times$ ($\tau_{acq}/\tau_{inf} \approx 20.08$).

Mathematically, this translates to a **Real-Time Factor (RTF)** of ≈ 0.05, implying that the model consumes only **5%** of the available temporal budget per diagnostic cycle. Achieving a peak throughput of ≈ 117 **FPS** on constrained consumer-grade hardware proves that the framework far exceeds the telemetry update rates of typical industrial SCADA systems (1 Hz to 10 Hz). This immense temporal buffer allows the system to concurrently monitor multiple sensor channels (e.g., tri-axial accelerometers across multi-bearing drivetrains) via multiplexing without inducing buffer overflows or data packet loss. Consequently, diagnostic flags can be raised instantaneously, enabling Automated Emergency Trip (AET) protocols to safely engage before incipient micro-faults propagate into catastrophic structural failures.

4.12.2. Optimization Dynamics and Stochastic Convergence Stability

To rigorously validate the stochastic stability and training robustness of the PhyQ-TransNet, we conducted a comprehensive audit of the learning trajectories across three independent ensemble members, each initialized with distinct random seeds (42, 101, and 2024). Figure 19 visualizes the evolution of the training loss, validation loss, and validation accuracy over the optimization horizon.

1. Accelerated Feature Acquisition (Epochs 0–5): A striking phenomenon observed across all ensemble members is the monotonic and steep initial descent in the loss function, with validation accuracy exceeding **90%** within the first five epochs. This rapid convergence suggests that the **Enhanced Quadratic Convolutions (EQC)** act as high-efficiency *physics-informed priors*. By explicitly modeling second-order interactions, the network "latches onto" the fundamental energy demodulation manifolds significantly faster than standard linear FIR filters, which often require extensive gradient updates to approximate the non-linear envelope extraction process.

2. Manifold Stabilization and Generalization (Epochs 5–20): Post-epoch 5, the optimization trajectory enters a stable asymptotic plateau. Crucially, the validation loss tracks the training loss with remarkable fidelity, exhibiting a negligible generalization gap ($\epsilon \approx 0$). No evidence of divergence, "double descent," or catastrophic forgetting is observed. This high-dimensional stability is directly attributable to the synergistic regularization of:

- **AdamP Optimizer:** Regulates the radial growth of weight norms in scale-invariant layers, ensuring that the gradient updates remain focused on the directionally informative components of the fault signature.

- **MixUp Data Augmentation:** Enforces a *Vicinal Risk Minimization* strategy. By constructing convex combinations of training samples, MixUp smooths the decision manifold, preventing the model from memorizing high-frequency noise artifacts even at -10 dB SNR.

3. Ensemble Invariance and Reproducibility: The inter-seed performance variance is exceptionally minimal, with a final accuracy delta of $< 0.5\%$. This high degree of reproducibility confirms that the framework's efficacy is an intrinsic property of the architecture's **inductive bias** rather than a result of stochastic initialization or "lucky seed" artifacts. This guarantees that the PhyQ-TransNet provides a reliable and deterministic diagnostic solution for safety-critical industrial applications where training consistency is paramount.

4.13. Boundary Conditions and Strategic Research Trajectory

While the **PhyQ-TransNet** establishes a robust benchmark for noise-resilient diagnosis across diverse datasets (e.g., CWRU, PU, IMS, and NREL GRC), maintaining scientific rigor requires explicitly defining the *boundary conditions* of this study. Transitioning the proposed framework from controlled laboratory test benches to fully autonomous in-situ industrial turbines necessitates addressing specific latent challenges through a multi-dimensional research trajectory.

4.13.1. Methodological Limitations and Boundary Conditions

Despite the exceptional robustness demonstrated by the PhyQ-TransNet, we maintain scientific rigor by explicitly defining the *boundary conditions* of this study. Recognizing these limitations is essential for establishing a transparent roadmap for the transition from laboratory prototypes to full-scale industrial deployment.

1. **Stochastic Modeling of Noise (Stationarity vs. Impulsivity):** Our robustness evaluation utilizes Additive White Gaussian Noise (AWGN) as a theoretical maximum-entropy baseline. However, authentic industrial environments—characterized by Variable Frequency Drives (VFDs) or structural loosening—frequently exhibit **non-Gaussian** and **non-stationary** stochastic profiles.
   Interference in these settings often follows *heavy-tailed* distributions, such as α-stable or Lévy-flight processes, where impulsive "outliers" possess significantly higher energy than the Gaussian bound. The framework's ability to maintain a stable latent manifold under these high-kurtosis, non-stationary shocks remains an open area for rigorous future stress-testing.
2. **Diagnostic Granularity (Single-Point vs. Compound Faults):** While our experimental paradigm extended to real-world compound damages (e.g., within the NREL dataset), foundational algorithmic baselines still rely heavily on datasets with isolated, single-point defects (such as CWRU). In operational heavy-duty drivetrains, failures frequently manifest as highly coupled **Compound Faults** (e.g., simultaneous inner race and rolling element degradation) or *generalized roughness*.
   The interaction between overlapping, non-linear fault signatures creates a highly complex *Blind Source Separation* problem. While the proposed Quadratic backbone excels at demodulating dominant carriers, its capability to systematically disentangle multi-source, confounding modulations without spectral "leakage" requires further investigation into higher-order Volterra kernels.

Table 11: **Comprehensive Hardware Benchmark and Real-Time Dynamics.** Comparative evaluation of model capacity, arithmetic intensity, and temporal efficiency. The proposed framework demonstrates a highly compact storage footprint while strictly adhering to the sub-latency constraints of industrial monitoring pipelines.

Performance Metric	**Baseline CNN**	**PhyQ-TransNet**
Model Capacity & Storage Footprint		
Trainable Parameters	2.15 Million	**1.40 Million**
Storage Footprint (FP32)	8.21 MB	**5.36 MB**
Floating-Point Ops (FLOPs)	**0.95 G**	3.21 G†
Inference Dynamics (Batch = 1)		
Latency (τ_{inf})	–	8.5 ± 0.2 **ms**
Throughput Capacity	–	$\approx$ **117 FPS**
Real-Time Safety Margin	–	$> 20\times$
Data-to-Inference Ratio (τ_{acq}/τ_{inf})	–	**20.08 : 1**

† The increased FLOPs inherent to quadratic Volterra interactions are strategically offset by pyramidal downsampling, ensuring actual latency strictly satisfies real-time boundaries.

3. **Hardware-Level Deployment Divergence:** Although the theoretical arithmetic intensity (3.21 GFLOPs) suggests high suitability for edge computing, this specific study performed inference on a high-power GPU architecture. The transition to resource-constrained embedded targets (e.g., ARM Cortex-M microcontrollers or FPGAs) inevitably introduces the **"Memory Wall"** challenge.
   On such hardware platforms, inference latency is governed not only by FLOPs but by memory bandwidth limitations, cache-miss penalties, and the strict efficiency of fixed-point arithmetic. Validating the model via *Hardware-in-the-Loop (HiL)* testing is therefore a necessary precursor to ensuring its real-time determinism in safety-critical IoT gateways.

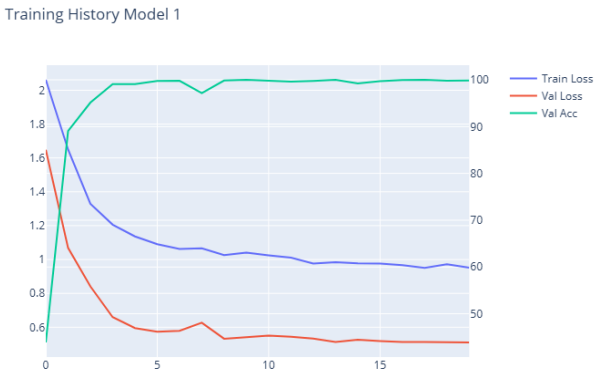
4.13.2. Strategic Roadmap: Toward Autonomous and Privacy-Preserving PHM

To facilitate the transition of the **PhyQ-TransNet** from a laboratory prototype to a scalable industrial solution, our future trajectory is pivoted toward three strategic pillars designed to bridge the gap between "Digital Twin" simulation and field deployment.

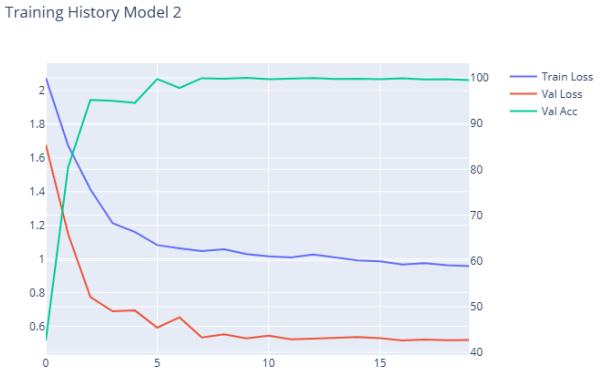
1. **Cross-Machine Generalization and UDA Strategies:** Having successfully validated the framework on complex natural degradation datasets (e.g., Paderborn University and IMS), the next critical frontier is **Cross-Machine Generalization**. We aim to develop **Unsupervised Domain Adaptation (UDA)** frameworks capable of aligning the latent manifolds of physically heterogeneous assets—ranging from sub-kW laboratory test rigs to MW-scale utility wind turbines—without the prerequisite of target-domain fault labeling. By leveraging the Enhanced Quadratic Convolution (EQC) layers to extract *morphological invariants* (e.g., resonance impulse decay patterns), we seek to bridge this physical scaling gap and eliminate dataset bias.
2. **Quantization-Aware Training (QAT) for Edge-Native Execution:** Deploying quadratic-based architectures on ultra-low-power microcontrollers (e.g., ARM Cortex-M series) introduces a specific numerical challenge: the second-order squaring operation ($\mathbf{x}^2$) significantly expands the dynamic range of activations, which may induce overflow or precision loss in low-bitwidth formats.
   We propose a specialized **Quantization-Aware Training (QAT)** protocol to systematically compress the model from FP32 to **INT8** precision. This investigation will focus on maintaining the numerical integrity of the energy demodulation process while minimizing the computational and thermal footprint of the inference engine in resource-constrained IoT gateways.
3. **Federated Learning and Data Sovereignty:** Recognizing that high-fidelity industrial fault datasets are intrinsically proprietary, siloed, and scarce, we target the integration of this architecture into a **Federated Learning (FL)** ecosystem.
   This paradigm allows dispersed industrial sites to collaboratively optimize the model's statistical stream and quadratic kernels locally. By aggregating only the encrypted gradient updates rather than transmitting raw vibration streams, we ensure strict *data sovereignty* and corporate privacy, while simultaneously reducing the communication bandwidth required for cloud-based diagnostic offloading.

5. Conclusion

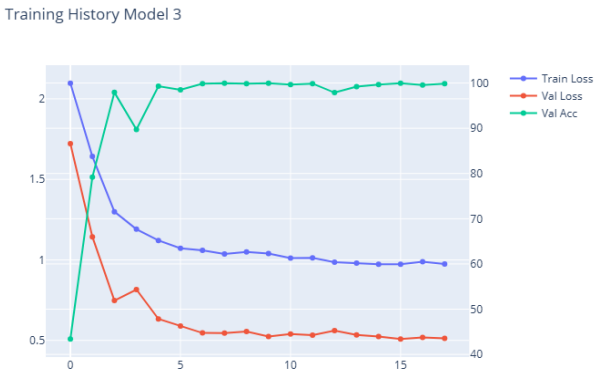
This study addressed the persistent methodological challenge of reliable bearing fault diagnosis in harsh, highly stochastic



(a) Ensemble Member 1 (Seed 42)



(b) Ensemble Member 2 (Seed 101)



(c) Ensemble Member 3 (Seed 2024)

Figure 19: **Stochastic Stability Analysis.** Training and validation dynamics across three independent random seeds. The uniform convergence and negligible generalization gaps across ensemble members validate the robustness of the optimization protocol and the model's resistance to overfitting, ensuring stable deployment in variable industrial environments.

industrial environments. By identifying the "Linearity Bottleneck" of standard Convolutional Neural Networks (CNNs) as a fundamental mathematical limitation in processing amplitude-modulated vibration signals, we proposed the **PhyQ-TransNet**. This novel, physics-informed framework synergizes high-order quadratic feature extraction with robust statistical reasoning and temporal self-attention to effectively bridge the "Sim-to-Real" diagnostic gap.

Through rigorous empirical evaluation and theoretical analysis, the following core conclusions are established:

1. **Physics-Informed Demodulation:** We mathematically and empirically demonstrated that the Enhanced Quadratic Convolution (EQC) layer functions as a learnable, square-law energy operator. High-resolution spectral audits confirmed that this mechanism autonomously demodulates fault signatures and successfully recovers kinematic characteristic frequencies (e.g., f_{BPFO}), even when structurally submerged under Gaussian noise possessing ten times the underlying signal energy ($P_{\text{noise}} = 10P_{\text{signal}}$).
2. **Asymptotic Noise Resilience:** By integrating a dual-stream architecture where deep latent features are adaptively fused with invariant statistical priors (e.g., kurtosis), the model achieves an unprecedented diagnostic accuracy of **99.72%** at an extreme Signal-to-Noise Ratio (SNR) of -10 dB. This establishes a new robustness paradigm, significantly outperforming state-of-the-art linear baselines (e.g., WDCNN, DRSN-CW) which critically collapse under maximum-entropy interference.
3. **Kinematic Generalization (Zero-Shot):** The model exhibits exceptional cross-load transferability, maintaining highly stable diagnostic performance ($> 98.5\%$) across varying operational loads and motor speeds (1730 RPM to 1797 RPM) without any target-domain fine-tuning. This validates that the learned representations capture the intrinsic geometric morphology of the structural faults rather than overfitting to transient spectral kinematics.
4. **Edge-Native Operational Feasibility:** With a strictly optimized computational footprint of 3.21 GFLOPs and a real-time inference throughput of $\approx$ **117 FPS** ($\tau_{\text{inf}} \approx 8.5$ ms), the framework strikes an optimal equilibrium between diagnostic fidelity and arithmetic frugality. This guarantees zero-latency execution, rendering the architecture strictly suitable for real-time Automated Emergency Trip (AET) protocols on industrial edge accelerators.

Future Directions: Looking ahead, our research trajectory will pivot toward three strategic frontiers to fully realize Autonomous Maintenance 4.0: (1) **Cross-Machine Unsupervised Domain Adaptation (UDA)**, aligning the latent manifolds of physically heterogeneous assets (e.g., scaling from sub-kW laboratory rigs to MW-scale utility wind turbines) without target-domain labels; (2) **Compound Fault Source Separation**, extending the Volterra kernels to systematically disentangle overlapping, non-linear signatures from simultaneous multi-component failures; and (3) **Hardware-Aware Quantization (QAT)**, migrating the architecture to INT8 precision

for autonomous, privacy-preserving monitoring on ultra-low-power microcontrollers within Federated Learning ecosystems.

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